

Unit 1

Numbers

Here's a brief explanation of each concept:

1. **Complex Numbers:** Numbers that have both a real part and an imaginary part. A complex number is written in the form $a + bi$, where a is the real part, and bi is the imaginary part. i is the imaginary unit, defined as $i^2 = -1$.
2. **Imaginary Numbers:** Numbers that are multiples of the imaginary unit i , where $i = \sqrt{-1}$. Examples include $5i$ or $-3i$. These numbers are used when dealing with the square root of negative numbers, which are not possible within real numbers.
3. **Real Numbers:** All numbers that can be found on the number line, including both rational and irrational numbers. Real numbers can be positive, negative, or zero. Examples include -2 , 0 , 3.5 , $\sqrt{2}$.
4. **Rational Numbers:** Numbers that can be expressed as a ratio of two integers (i.e., fractions), where the denominator is not zero. For example, $\frac{1}{2}$, $-\frac{3}{4}$, 5 , and 0.75 are rational numbers.
5. **Irrational Numbers:** Numbers that cannot be expressed as a fraction of two integers. Their decimal expansion goes on forever without repeating. Examples include π , e , and $\sqrt{2}$.
6. **Integers:** Whole numbers and their negatives, including zero. Examples are -3 , -2 , -1 , 0 , 1 , 2 , 3 .
7. **Whole Numbers:** Non-negative integers, which include 0 and all positive integers. Examples are 0 , 1 , 2 , 3 , 4 , 5 .
8. **Natural Numbers:** The set of positive integers starting from 1 . Natural numbers are 1 , 2 , 3 , 4 , $\dots$. In some definitions, 0 is also included.
9. **Prime Numbers:** Natural numbers greater than 1 that have only two positive divisors: 1 and the number itself. Examples include 2 , 3 , 5 , 7 , 11 , 13 .
10. **Perfect Numbers:** A perfect number is a positive integer that is equal to the sum of its proper divisors (excluding itself). For example, 6 is a perfect number because its divisors (1 , 2 , 3) add up to 6 , and 28 is another example (divisors: 1 , 2 , 4 , 7 , 14).

Divisibility Rules

1. Divisibility by 2:

A number is divisible by 2 if its unit digit is even.

2. Divisibility by 3:

A number is divisible by 3 when the sum of its digits is divisible by 3.

3. Divisibility by 4:

A number is divisible by 4 when the number formed by its two extreme right digits (i.e., tens digit and unit digit) is divisible by 4.

4. Divisibility by 5:

A number is divisible by 5 when its unit digit is either 5 or 0.

5. Divisibility by 6:

A number is divisible by 6 when it is divisible by both 2 and 3.

6. Divisibility by 7:

Method A:

Write the given number.

Write the number leaving its extreme right digit and subtract from it the product of the extreme right digit and 2. The balance obtained is a new number.

Repeat step 2 until you can determine if the balance is divisible by 7 or not.

Method B (For numbers with more than 3 digits):

Group the number into groups of three, starting from the unit digit.

Add the sums of the odd groups and even groups separately.

The difference between the sum of the odd and even groups should be divisible by 7.

7. Divisibility by 8:

A number is divisible by 8 when the number formed by its three extreme right digits (i.e., hundreds digit, tens digit, and unit digit) is divisible by 8.

8. Divisibility by 9:

A number is divisible by 9 when the sum of its digits is divisible by 9.

9. Divisibility by 10:

A number is divisible by 10 when it has zero (0) in its unit's place.

10. Divisibility by 11:

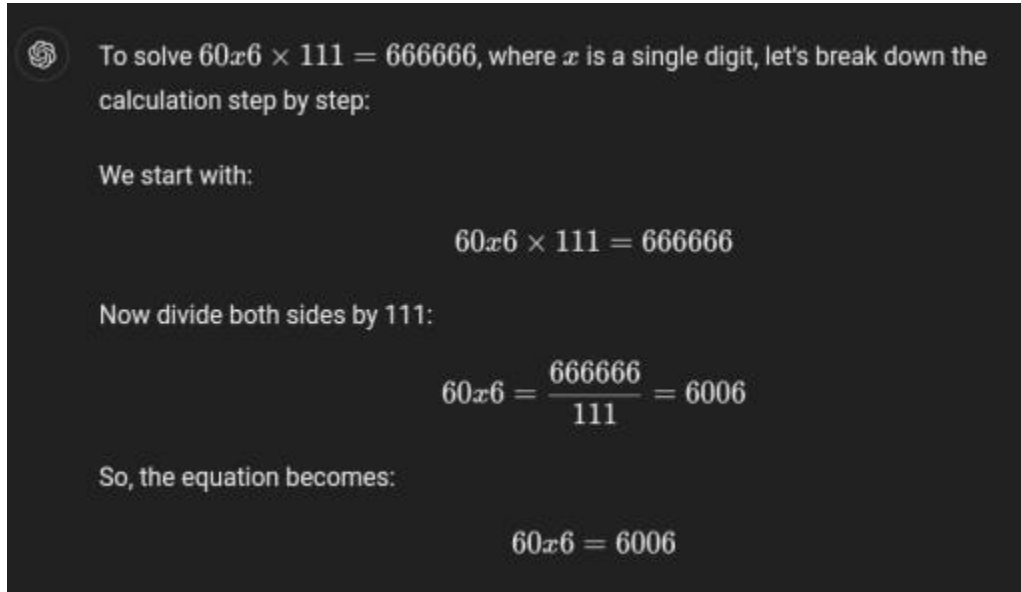
A number is divisible by 11 when the difference between the sum of the digits at odd places and the sum of the digits at even places is either zero (0) or divisible by 11.

11. Divisibility by 12:

A number is divisible by 12 when it is divisible by both 3 and 4.

Sums on Divisibility

1. $60x6 \times 111 = 666666$, find x .



The image is a screenshot of a chatbot interface with a dark background. It shows a step-by-step solution to the equation $60x6 \times 111 = 666666$. The chatbot's avatar is a small circular icon with a spiral pattern. The text is white and follows a logical sequence: stating the problem, breaking it down, starting with the equation, dividing both sides by 111, and simplifying to $60x6 = 6006$.

To solve $60x6 \times 111 = 666666$, where x is a single digit, let's break down the calculation step by step:

We start with:

$$60x6 \times 111 = 666666$$

Now divide both sides by 111:

$$60x6 = \frac{666666}{111} = 6006$$

So, the equation becomes:

$$60x6 = 6006$$

Therefore $x = 0$

2. If $23334x$ is divisible by 5, Find number of possible values of x .

Ans: As the number is divisible by 5, the units place would be either 0 or 5, hence the number of values are 2.

3. 24 is divided into two parts such that 7 times the first part added to 5 times the second part makes 146. What is the first part?

Ans: let the first part be x and the second part be $24-x$

According to the problem, 7 times the first part added to 5 times the second part equals 146:

$$\begin{aligned}7x + 5(24-x) &= 146 \\7x + 120 - 5x &= 146 \\2x + 120 &= 146 \\2x &= 146 - 120 \\2x &= 26 \\x &= \frac{26}{2} \\x &= 13\end{aligned}$$

∴ The first part is 13

For divisibility sums like : "517*324 is divisible by 3 find what is *?"

There are two approaches to solving such a problem. Either you remember the divisibility rules for numbers from 1 to 12, or you can use brute force with a calculator. For example:

Replace * with 1. If the number is divisible by 3, then * = 1 is the answer. If not, replace * with 2, then with 3, 4, ..., 9, and 0 until you find that the number is divisible by 3.

Sums on H.C.F

1. Find HCF of 70 and 90

1. Find HCF of 70 and 90

Ans: Step 1:

2 70	2 90
5 35	5 45
7 7	3 9
1	3 3
	1

$2 \times 5 \times 7$ $2 \times 5 \times 3 \times 3$

Step 2: Get common factors from both numbers

$(2) \times (5) \times 7$
 $(2) \times (5) \times 3 \times 3$

$2 \times 5 = 10$

∴ H.C.F of 70 and 90 is 10

2. Find HCF of $\frac{2}{5}$, $\frac{3}{10}$, $\frac{6}{25}$

2 What is the HCF of $\frac{2}{5}$, $\frac{3}{10}$, $\frac{6}{25}$

To find HCF of such numbers we have to find H.C.F of numerators and L.C.M of denominators

Step 1: Separate the numbers

$2, 3, 6 \rightarrow$ numerators
 $5, 10, 25 \rightarrow$ denominators

Step 2: Find H.C.F of numerators

2	2	3	3	2	6
1	1	1	1	3	3
1	1	1	1	1	1

Step 3: Multiply the common multiples which appear in all the numbers

$2 \times \textcircled{1}$
 $3 \times \textcircled{1} = 1$
 $2 \times 3 \times \textcircled{1}$

We have got the H.C.F of all the numerators as 1

Step 4: Find the L.C.M of denominators

5	5	2	10	5	25
1	1	5	5	5	5
1	1	1	1	1	1

Step 5: Multiply the common and uncommon multiples

$\textcircled{5} \times \textcircled{1}$
 $2 \times \textcircled{5} \times \textcircled{1} = 5 \times 1 \times 2 \times 5$
 $\textcircled{5} \times 5 \times \textcircled{1} = 50$

We have got the L.C.M of denominators as 50

$\therefore \text{H.C.F of } \frac{2}{5}, \frac{3}{10}, \frac{6}{25} = \frac{1}{50}$

Sums on L.C.M

1. Find L.C.M of 25 and 30

1. Find LCM of 25 and 30

Ans. Step 1:

5		25
5		5
		1

2		30
3		15
5		5
		1

Step 2: Put together the common multiples

$$\begin{array}{l} \textcircled{5} \times 5 \\ 2 \times 3 \times \textcircled{5} \end{array} = 5$$

Step 3: Add the remaining numbers to the equation

$$\begin{array}{l} \textcircled{5} \times 5 \\ 2 \times 3 \times \textcircled{5} \end{array} = 5 \times 5 \times 2 \times 3$$
$$= 150$$

$\therefore$ LCM of 25 and 30 is 150

2. What is the L.C.M of $2/5$, $3/10$, $6/25$

2 What is the LCM of $2/5$, $3/10$, $6/25$

Ans: To find LCM of such numbers we have to find L.C.M of numerators and H.C.F of denominators

Step I: Separate the numbers

2, 3, 6 $\rightarrow$ numerators

5, 10, 25 $\rightarrow$ Denominators

Step 2: Find L.C.M of numerators

$$\begin{array}{r|l} 2 & 2 \\ 1 & 1 \\ & 1 \end{array}$$

$$\begin{array}{r|l} 3 & 3 \\ 1 & 1 \\ & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 6 \\ 3 & 3 \\ 1 & 1 \end{array}$$

Step 3: Multiply common and uncommon multiples

$$\begin{aligned} & \textcircled{2} \times \textcircled{1} \\ & \textcircled{3} \times \textcircled{1} \\ & \textcircled{2} \times \textcircled{3} \times \textcircled{1} \end{aligned} = 1 \times 3 \times 2 = 6$$

We got the L.C.M of numerators as 6

Step 4: Now we calculate the H.C.F of denominators

$$\begin{array}{r|l} 5 & 5 \\ 1 & 1 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 10 \\ 5 & 5 \\ 1 & 1 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 5 & 25 \\ 5 & 5 \\ 1 & 1 \\ \hline & 1 \end{array}$$

Step 5: Multiply the common multiples

$$(5 \times 1)$$

$$2 \times (5) \times (1) = 5 \times 1$$

$$(5 \times 5 \times 1)$$

$$= 5$$

We got the H.C.F of denominators as 5

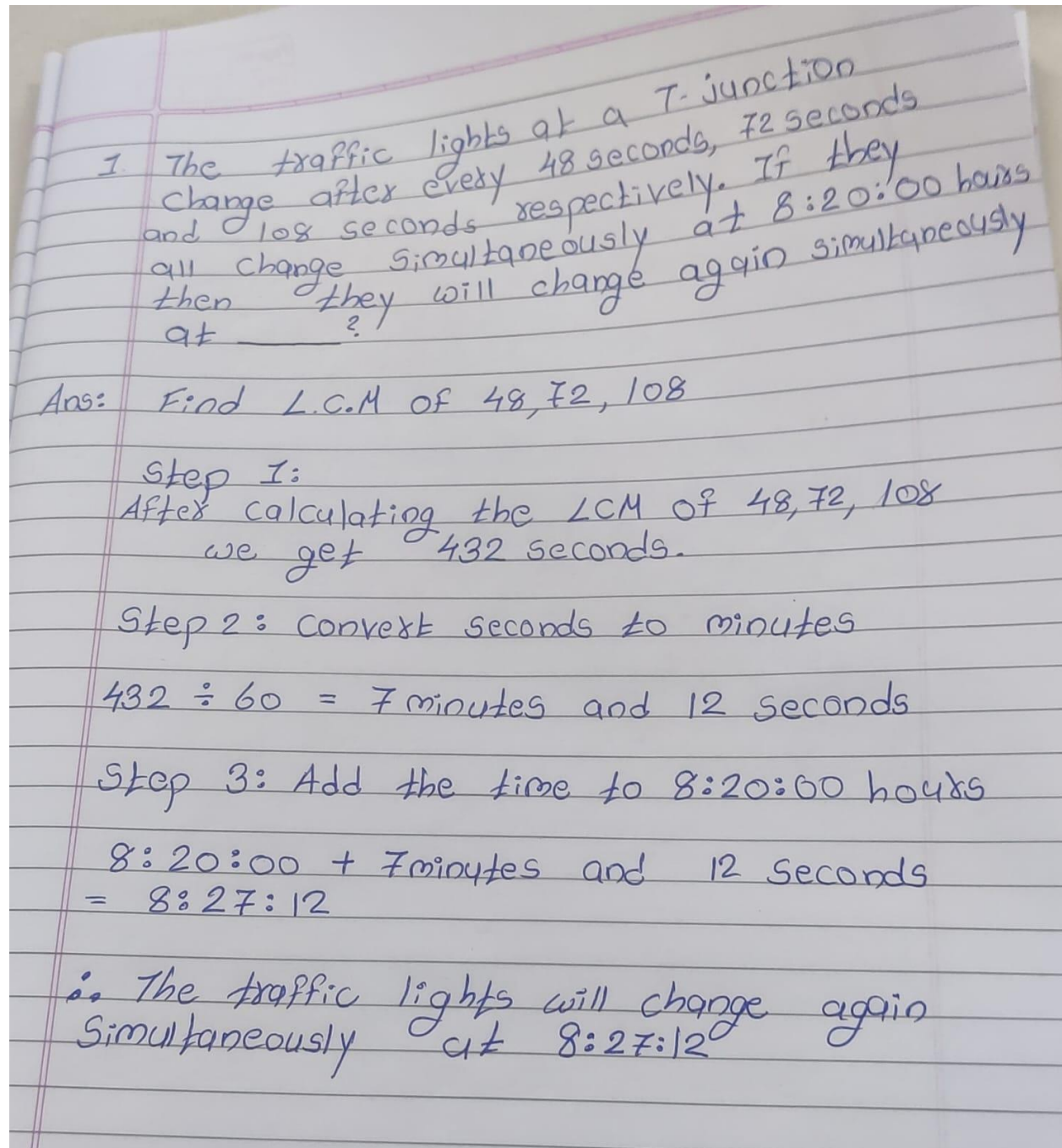
$\therefore$ The LCM of $2/5$, $3/10$, $6/25$

is

$$\frac{6}{5}$$

Word problems on H.C.F and L.C.M

1. The traffic lights at a T-junction change after every 48 sec, 72 sec and 108 sec respectively. If they all change simultaneously at 8:20:00 hours, then they will again change simultaneously at:



2. An oil merchant has three different kinds of oil: 435 liters, 493 liters and 551 liters. Find the least number of casks of equal size required to store all the oil without mixing

1. An oil merchant has three different kinds of oil: 435 l, 493 l, 551 l. Find the least number of casks of equal size required to store all the oil without mixing.

Ans: We have to first find the greatest size of a cask (oil barrel)

Step 1: Find H.C.F of 435, 493 and 551

3	435	17	493	19	551
5	145	29	29	29	29
29	29		1		1
	1				

After calculating we get the H.C.F as 29

Step 2: Divide oil quantities by H.C.F (29)

- $435 \div 29 = 15$
- $493 \div 29 = 17$
- $551 \div 29 = 19$

Step 3: Add all the numbers

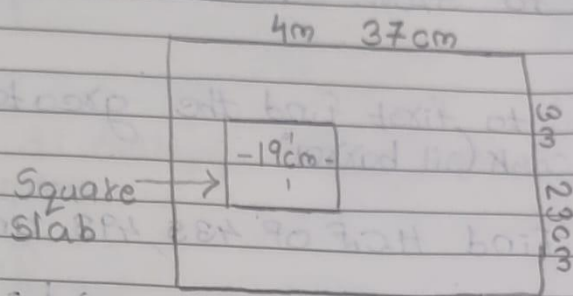
$$15 + 17 + 19 = 51$$

$\therefore$ The least number of casks required to store oil without mixing is 51

3. A room is 4 meters 37 cm long and 3 meters 23 cm broad. It is required to pave the floor with minimum square slabs. Find the number of slabs required for this purpose.

2. A room is 4m 37cm long and 3m 23cm broad. Find the minimum number of slabs required to pave the floor.

Ans:



Step I: Convert measurements to cm for convenience

- 4m 37cm = 437cm
- 3m 23cm = 323cm

Step 2: Find HCF of 437 and 323

The H.C.F of 437 and 323 is 19cm

Step 3: Calculate the number of square slabs

- Slabs required along the length = $437 \div 19 = 23$
- Slabs required along the breadth = $323 \div 19 = 17$

Step 4: Add the total slabs

$23 + 17 = 40$

$\therefore$ Number of square slabs required = 40

Note: when to Use LCM or HCF in word problems:

LCM:

- You need to find something that happens repeatedly after a certain interval and want to know when it will occur together again.
- The task is to find the smallest time or number where events or items coincide or match.

HCF:

- You need to divide something into the largest possible equal parts.
 - The task is to distribute items or arrange things such that the groups or arrangements are equal and as large as possible.
-

Cyclicity

Cyclicity of numbers refers to the repeating pattern in the units digit of the powers of a number. In simpler terms, when you raise a number to successive powers, the last digit of the result tends to follow a cyclical pattern. This property is especially useful in modular arithmetic and number theory, particularly when calculating large powers without directly multiplying large numbers.

Example of Cyclicity:

Consider the number 2:

- $2^1 = 2$ (units digit is 2)
- $2^2 = 4$ (units digit is 4)
- $2^3 = 8$ (units digit is 8)
- $2^4 = 16$ (units digit is 6)
- $2^5 = 32$ (units digit is 2)
- $2^6 = 64$ (units digit is 4)

You can see that the units digits (2, 4, 8, 6) repeat every 4 terms, so the cyclicity of 2 is 4.

Cyclicity Table

Number	Cyclicity	Power Cycle
1	1	1
2	4	2, 4, 8, 6
3	4	3, 9, 7, 1
4	2	4, 6
5	1	5
6	1	6
7	4	7, 9, 3, 1
8	4	8, 4, 2, 6
9	2	9, 1
10	1	0

Note: For calculating sums faster it is better to memorize this table

Sums on Cyclicity

1. Find the Unit digit of 287^{562581}

To find the unit digit of 287^{562581} , we focus only on the last digit of 287, which is 7.

The unit digit of powers of 7 follows a cycle of 4:

$$\begin{aligned} 7^1 &= 7 \quad (\text{unit digit: } 7) \\ 7^2 &= 49 \quad (\text{unit digit: } 9) \\ 7^3 &= 343 \quad (\text{unit digit: } 3) \\ 7^4 &= 2401 \quad (\text{unit digit: } 1) \end{aligned}$$

The pattern repeats every 4 powers: 7, 9, 3, 1.

To find the unit digit of 287^{562581} , we only need to calculate $562581 \bmod 4$:

$$562581 \div 4 = 140645 \text{ remainder } 1$$

Since the remainder is 1, the unit digit of 287^{562581} will be the same as the unit digit of 7^1 , which is 7.

Thus, the unit digit of 287^{562581} is 7.

2. Find the Unit digit of 13445×54336

To find the unit digit of 13445×54336 , we can focus on the unit digits of both numbers.

- The unit digit of 13445 is 5.
- The unit digit of 54336 is 6.

Now, multiply the unit digits:

$$5 \times 6 = 30$$

The unit digit of 30 is 0.

Thus, the unit digit of 13445×54336 is 0.

3. Find the last digit of $3119^{234} \times 234 \times 3119$

To find the last digit of $3119^{234} \times 234 \times 3119$, we can break it into two steps:

1. Find the last digit of 3119^{234} :

- The last digit of 3119 is 9, so we only need to consider 9^{234} .
- Powers of 9 follow a cycle of 2:

$$\begin{aligned} 9^1 &= 9 && \text{(last digit: 9)} \\ 9^2 &= 81 && \text{(last digit: 1)} \\ 9^3 &= 729 && \text{(last digit: 9)} \\ 9^4 &= 6561 && \text{(last digit: 1)} \end{aligned}$$

The pattern alternates between 9 and 1.

- Since $234 \div 2 = 117$, which is even, the last digit of 9^{234} is 1.

2. Find the last digit of 234×3119 :

- The last digit of 234 is 4 and the last digit of 3119 is 9.
- Now, multiply the last digits:

$$4 \times 9 = 36$$

The last digit of 36 is 6.

Now, multiplying the results of both parts:

- The last digit of 3119^{234} is 1.
- The last digit of 234×3119 is 6.

Finally, $1 \times 6 = 6$.

Thus, the last digit of $3119^{234} \times 234 \times 3119$ is 6.

4. Find the unit digit of the expression: $888^{888!} + 222^{222!} + 333^{333!} + 777^{777!}$

To find the unit digit of the expression:

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we will analyze the unit digits of each individual term separately, considering only the last digits of the bases (since the last digits of powers form cycles). Here's the breakdown:

1. Unit digit of $888^{888!}$:

The last digit of 888 is 8, so we focus on powers of 8:

- Powers of 8 follow a cycle of 4:

$8^1 = 8$	(unit digit: 8)
$8^2 = 64$	(unit digit: 4)
$8^3 = 512$	(unit digit: 2)
$8^4 = 4096$	(unit digit: 6)

The pattern repeats every 4 powers: 8, 4, 2, 6.

Since $888!$ is a multiple of 4, the unit digit of $888^{888!}$ is the same as 8^4 , which is 6.

2. Unit digit of $222^{222!}$:

The last digit of 222 is 2, so we consider powers of 2:

- Powers of 2 follow a cycle of 4:

$2^1 = 2$	(unit digit: 2)
$2^2 = 4$	(unit digit: 4)
$2^3 = 8$	(unit digit: 8)
$2^4 = 16$	(unit digit: 6)

Since $222!$ is a multiple of 4, the unit digit of $222^{222!}$ is the same as 2^4 , which is 6.

3. Unit digit of $333^{333!}$:

The last digit of 333 is 3, so we consider powers of 3:

- Powers of 3 follow a cycle of 4:

$3^1 = 3$	(unit digit: 3)
$3^2 = 9$	(unit digit: 9)
$3^3 = 27$	(unit digit: 7)
$3^4 = 81$	(unit digit: 1)

Since $333!$ is a multiple of 4, the unit digit of $333^{333!}$ is the same as 3^4 , which is 1.

4. Unit digit of $777^{777!}$:

The last digit of 777 is 7, so we consider powers of 7:

- Powers of 7 follow a cycle of 4:

$7^1 = 7$	(unit digit: 7)
$7^2 = 49$	(unit digit: 9)
$7^3 = 343$	(unit digit: 3)
$7^4 = 2401$	(unit digit: 1)

Since $777!$ is a multiple of 4, the unit digit of $777^{777!}$ is the same as 7^4 , which is 1.

Final Calculation:

Now, we sum the unit digits of the four terms:

$$6 + 6 + 1 + 1 = 14$$

The unit digit of 14 is 4.

Thus, the unit digit of the expression $888^{888!} + 222^{222!} + 333^{333!} + 777^{777!}$ is 4.

Unit 2

Fraction to Percentage

To convert a fraction to a percentage, follow these steps:

1. Divide the numerator (top number) by the denominator (bottom number) to get a decimal.
2. Multiply the decimal by 100 to get the percentage.

Example:

Convert $\frac{3}{4}$ to a percentage

- $3 \div 4 = 0.753 \times 100 = 75.3\%$
-

Percentage to Fraction

Example:

Convert 60% to a fraction:

- $60\% = \frac{60}{100}$
 - Simplify $\frac{60}{100} = \frac{3}{5}$
-

Percentage Increase or Decrease of a Quantity

Example 1: Find increase in percentage from 30 to 189

Handwritten calculation for Example 1:

$$\begin{aligned}\text{Percentage increase} &= \frac{\text{New value} - \text{Original value}}{\text{Original value}} \times 100 \\ &= \frac{189 - 30}{30} \times 100 \\ &= \frac{159}{30} \times 100 \\ &= 5.3 \times 100 \\ &= 530\% \\ \therefore \text{Increase in percent} &= 530\%\end{aligned}$$

Example 2: Find decrease in percentage from 200 to 17

Handwritten calculation for Example 2:

$$\begin{aligned}\text{Percentage decrease} &= \frac{\text{Original value} - \text{New value}}{\text{Original value}} \times 100 \\ &= \frac{200 - 17}{200} \times 100 \\ &= 0.915 \times 100 \\ &= 91.5 \\ \therefore \text{Decrease in percentage} &= 91.5\%\end{aligned}$$

Sums on ratios and Fractions

1. A shopkeeper bought 600 oranges and 400 bananas. He found 15% of oranges and 8% of bananas were rotten. Find the percentage of fruits in good condition.

Ans: Total oranges = 600
Total bananas = 400
Total fruits = 1000

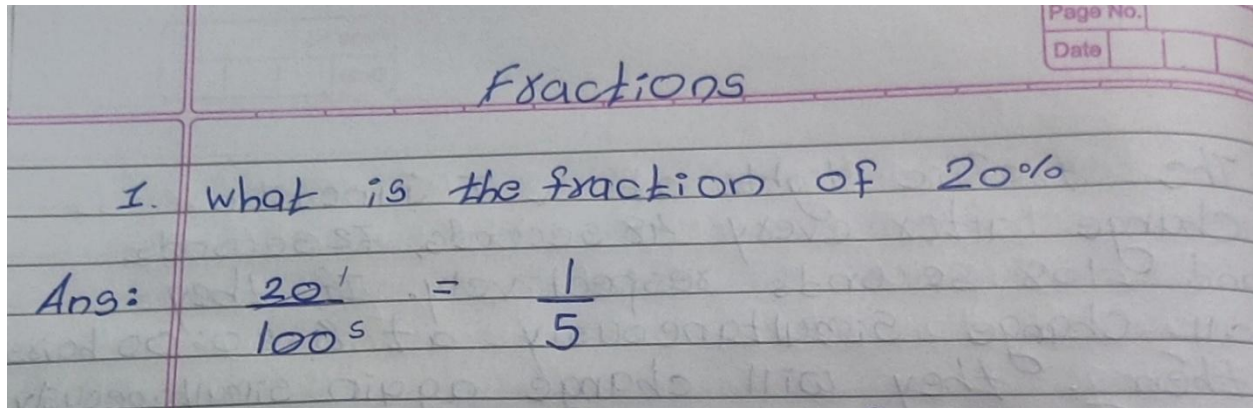
$$\begin{aligned}\text{Rotten oranges} &= 15\% \text{ of } 600 \\ &= \frac{15 \times 600}{100} \\ &= 90\end{aligned}$$

$$\begin{aligned}\text{Rotten bananas} &= 8\% \text{ of } 400 \\ &= \frac{8 \times 400}{100} \\ &= 32\end{aligned}$$

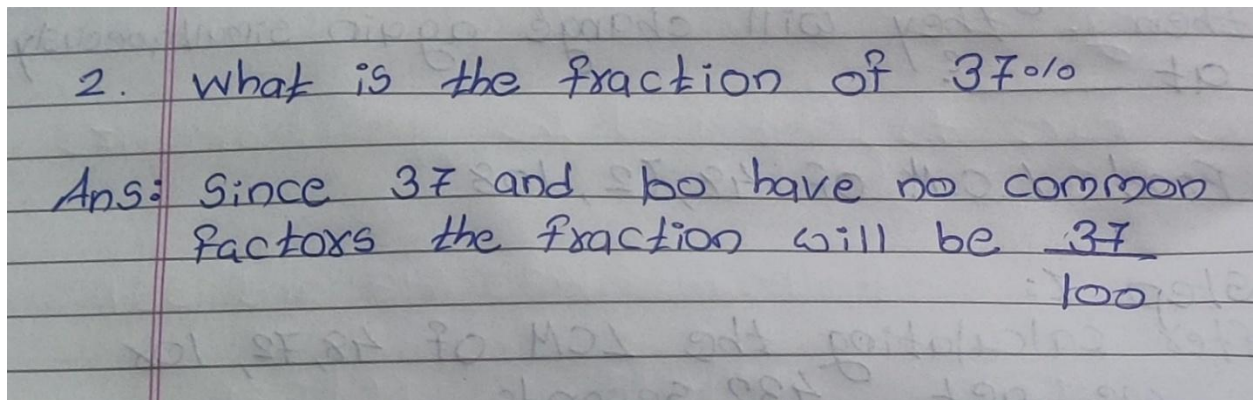
$$\text{Total rotten fruits} = 90 + 32 = 122$$

$$\begin{aligned}\therefore \text{Fruits in good condition} &= 1000 - 122 \\ &= 878 \\ &= 87.8\%\end{aligned}$$

2. What will be the fraction of 20 %



3. What is the fraction of 37%



4. In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was?

1. In an election between two candidates 'Candidate A' got 55% of total votes. 20% of votes were invalid. If total number of votes was 7500, the number of valid votes 'candidate B' got was?

Ans: Total votes = 7500
Invalid votes = 20%
∴ Total valid votes = 80% of 7500
= 6000

Candidate A got 55% of valid votes
which is 55% of 6000 = $\frac{55 \times 6000}{100}$
= 3300

Candidate B got the remaining votes which
is Total votes - candidate A votes
= 6000 - 3300
= 2700

5. Two students appeared at an examination. One of them secured 9 marks more than the other and his marks was 56% of the sum of their marks. The marks obtained by them are:

Ans: Let marks obtained by the two students be x and y . $x > y$

1. $x = y + 9$ (one student secured 9 marks more than the other).

2. $x = 0.56 \times (x + y)$ (his marks were 56% of the sum of their marks)

Now we substitute x from equation one to equation two.

$$\begin{aligned}y + 9 &= 0.56 \times ((y + 9) + y) \\&= y + 9 = 0.56 \times (2y + 9) \\&= y + 9 = 1.12y + 5.04 \\&= y + 9 - 5.04 = 1.12y \\&= y + 3.96 = 1.12y \\&= 3.96 = 1.12y - y \\&= 3.96 = 0.12y \\&\therefore y = \frac{3.96}{0.12} = 33\end{aligned}$$

Now substituting $y = 33$ in equation 1 to find x

$$\begin{aligned}x &= y + 9 \\x &= 33 + 9 \\x &= 42\end{aligned}$$

∴ One student scored 33 marks and the other scored 42

6. A man spends 35% of his income on food, 25% on children's education and 80% of the remaining on house rent. What percent of his income he is left with?

Ans: Let's assume his total income is 100%

1. Spending on food: 35%

$$100\% - 35\% = 65\%$$

2. Spending on Children's Education: 25%

$$65\% - 25\% = 40\%$$

3. Spending on House Rent: 80%

$$\begin{aligned} & 80\% \text{ of } 40\% \\ & = 0.8 \times 40\% \\ & = 32\% \end{aligned}$$

4. Remaining income after house rent:

$$\begin{aligned} & 40\% - 32\% \\ & = 8\% \end{aligned}$$

7. The ratio 5:20 expressed as percent equals to

Ans: $\frac{5}{20} \times 100$

We will substitute 20 and 100 with each other.

$$\frac{5}{\cancel{20}^1} \times \overset{5}{\cancel{100}_5} = 5 \times 5 = 25$$

$\therefore$ 5:20 in percentage equals to 25%

Sums on percentages

1. What percentage is equivalent to 514?

a. 500% b. 625%

c. 600% d. 525%

Ans: we need to find the number which is the closest to 514 out of the given options.
Since 525 is the closest our answer will be 525.

2. What is 10% of 30% of 20% of 1000

Ans: Calculate 20% of 1000:

1. $20\% \times 1000 = \frac{20}{100} \times 1000 = 200$

2. Calculate 30% of 200:

$30\% \times 200 = \frac{30}{100} \times 200 = 60$

3. Calculate 10% of 60:

$10\% \times 60 = \frac{10}{100} \times 60 = 6$

∴ our Answer is 6

3. If the salary of a person is first increased by 15% and then decreased by 12%, what is the percent change in his salary?

Ans: Net Percentage change = $a + b + \frac{a \times b}{100}$

where:

$$a = 15$$

$$b = -12$$

Now substituting the values:

$$N.P.C = 15 + (-12) + \frac{15 \times (-12)}{100}$$

$$= 15 - 12 + \frac{(-180)}{100}$$

$$= 3 - 1.8$$

$$= 1.2\%$$

$\therefore$ The net percent change in salary is an increase of 1.2%.

4. In a certain Mathematics examination, there were 2000 students, out of which 900 were boys and the rest were girls. If 32% of the boys and 38% of the girls passed, find the total percentage of failed students.

$$\begin{aligned}\text{Ans: Total Students} &= 2000 \\ \text{Boys} &= 900 \\ \text{Girls} &= 2000 - 900 = 1100\end{aligned}$$

1. Boys who passed = 32% of 900:

$$\frac{32}{100} \times 900 = 288$$

2. Girls who passed = 38% of 1100:

$$\frac{38}{100} \times 1100 = 418$$

3. Total students who passed = $288 + 418 = 706$.

4. Total students who failed = $2000 - 706 = 1294$

5. Percentage of failed students:

$$\frac{1294}{2000} \times 100 = 64.7\%$$

$\therefore$ Total percentage of failed students
is 64.7%

5. A metal bar weighs 8.15 ounces. 93% of the bar is silver. How many ounces of silver are in the bar?

Ans: Amount of silver = 93% of 8.15

$$\frac{93}{100} \times 8.15$$
$$= 0.93 \times 8.15$$
$$= 7.5$$

∴ The bar contains 7.5 ounces of silver.

Profit and loss formulas

Key Formulas in Profit and Loss:

1. Profit (P)

$$P = \text{Selling Price (SP)} - \text{Cost Price (CP)}$$

- If $SP > CP$, then there is a profit.

2. Loss (L)

$$L = \text{Cost Price (CP)} - \text{Selling Price (SP)}$$

- If $CP > SP$, then there is a loss.

3. Profit Percentage

$$\text{Profit \%} = \left(\frac{\text{Profit}}{\text{Cost Price}} \right) \times 100$$

$$\text{Profit \%} = \left(\frac{SP - CP}{CP} \right) \times 100$$

4. Loss Percentage

$$\text{Loss \%} = \left(\frac{\text{Loss}}{\text{Cost Price}} \right) \times 100$$

$$\text{Loss \%} = \left(\frac{CP - SP}{CP} \right) \times 100$$

5. Selling Price (SP) when profit

$$SP = CP + \text{Profit}$$

$$SP = CP + \left(\frac{\text{Profit \%} \times CP}{100} \right)$$

6. Selling Price (SP) when loss

$$SP = CP - \text{Loss}$$

$$SP = CP - \left(\frac{\text{Loss \%} \times CP}{100} \right)$$

7. Cost Price (CP) when profit

$$CP = \frac{SP}{1 + \frac{\text{Profit \%}}{100}}$$

8. Cost Price (CP) when loss

$$CP = \frac{SP}{1 - \frac{\text{Loss \%}}{100}}$$

Sums on Profit and Loss

1. The cost price of a shirt is Rs.200 and selling price is Rs.250. Find the percent profit.

The image shows a handwritten solution on lined paper. It starts with the formula for profit: $\text{Ans: Profit} = \text{Selling price} - \text{Cost price}$. Then it substitutes the values: $= 250 - 200$, resulting in $= 50$. Next, it uses the formula for profit percentage: $\text{Profit percentage} = \frac{\text{Profit}}{\text{C.P} \times 100}$. It substitutes the values: $= \frac{50}{200 \times 100}$, which simplifies to $= 25$. Finally, it concludes: $\therefore \text{Profit percentage is } 25\%$.

$$\begin{aligned}\text{Ans: Profit} &= \text{Selling price} - \text{Cost price} \\ &= 250 - 200 \\ &= 50 \\ \text{Profit percentage} &= \frac{\text{Profit}}{\text{C.P} \times 100} \\ &= \frac{50}{200 \times 100} \\ &= 25 \\ \therefore \text{Profit percentage is } 25\%\end{aligned}$$

2. Sudesh sells a car to Suresh at a profit of 5% and Suresh sells it to Ramesh at a profit of 10%. Find the resultant profit per cent.

Ans: Let the C.P of the car for Sudesh be Rs. 100 (Assuming for simplicity).

1. Sudesh sells the car to Suresh at a profit of 5%.

Selling price for Sudesh = C.P + 5% of C.P

$$= 100 + \left(\frac{5}{100}\right) \times 100$$

$$= 100 + 5$$

$$= \text{Rs. } 105$$

2. Suresh sells the car to Ramesh at a profit of 10%

C.P of car for Suresh is = Rs. 105

Selling price for Suresh = C.P + 10% of C.P

$$= 105 + \left(\frac{10}{100}\right) \times 105$$

$$= 105 + 10.5$$

$$= 115.5$$

$$\begin{aligned} 3. \text{ Total profit} &= \text{Final Selling price} - \overset{\text{initial}}{\text{Cost price}} \\ &= 115.5 - 100 \\ &= 15.5 \end{aligned}$$

$\therefore$ Resultant profit percentage = 15.5%

3. What is the single discount equivalent to two successive discounts of 10% and 20%

$$\text{Ans: Single Discount} = a + b - \frac{a \times b}{100}$$

Where:

$$a = 10\%, \quad b = 20\%$$

$$\text{Single Discount} = 10 + 20 - \frac{10 \times 20}{100}$$

$$= 30 - \frac{200}{100}$$

$$= 30 - 2$$

$$= 28$$

∴ Single Discount is 28%

4. By selling 33 meters of cloth, a shopkeeper gains the price of 11 meters of cloth. What is his gain per cent?

Ans: Total C.P for 33 meters = $33x$

Total S.P for 33 meters = C.P of 33 meters
+ Gain (which is the price of 11 meters)

So, S.P for 33 meters = $33x + 11x = 44x$

Profit = S.P - C.P = $44x - 33x = 11x$

Profit percentage = $\frac{\text{Profit}}{\text{C.P} \times 100}$

= $\frac{11x}{33x} \times 100$

= 33.33

∴ The gain percentage is 33.33%

5. Two washing machines were sold for Rs. 12000. One was sold at 20% loss and the other at 20% profit. What was the net gain or net loss in percent?

Ans: It is an overall loss because one of the washing machine was sold at a loss percentage.

$$\text{Net Loss Percentage} = \frac{(\text{Loss Percentage})^2}{100}$$
$$= \frac{20^2}{100} = \frac{400}{100} = 4$$

∴ There is a net loss of 4% in the transaction.

6. If three dozen pencils were purchased for Rs. 72 and were sold at the price of Rs. 72 for two dozen. Find the percentage profit or loss.

Ans: 1. Cost Price (C.P):

Three dozen pencils = Rs. 72

Cost of one dozen pencils = $\frac{72}{3}$ = Rs. 24

2. Selling Price (S.P):

Two dozen pencils sold at Rs. 72

Selling Price of One dozen pencils = $\frac{72}{2}$ = Rs. 36

3. Profit:

Profit = S.P - C.P

$= 36 - 24$
 $= 12$

4. Profit Percentage:

$\frac{\text{Profit}}{\text{C.P}} \times 100$

$= \frac{12}{24} \times 100 = 50$

∴ Profit percentage is 50%

Unit 3

Simple Interest Formula

The formula for calculating Simple Interest (SI) is:

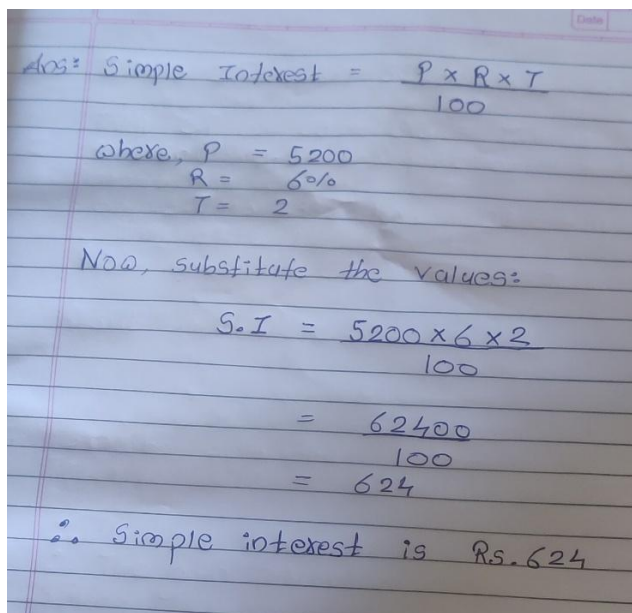
$$\text{Simple Interest (SI)} = \frac{\text{Principal (P)} \times \text{Rate (R)} \times \text{Time (T)}}{100}$$

where:

- P = Principal amount (initial amount of money)
- R = Rate of interest per year (as a percentage)
- T = Time the money is invested or borrowed for, in years

Simple Interest sums

1. Find the simple interest on Rs.5200 for two years at the rate of interest at 6% per annum.



Ans: Simple Interest = $\frac{P \times R \times T}{100}$

where, $P = 5200$
 $R = 6\%$
 $T = 2$

Now, substitute the values:

$$\begin{aligned} \text{S.I} &= \frac{5200 \times 6 \times 2}{100} \\ &= \frac{62400}{100} \\ &= 624 \end{aligned}$$

$\therefore$ Simple interest is Rs.624

2. A person borrows Rs.3000 from his friend at 15% per year for 3 years. What is the interest and the money to be returned respectively by the person to his friend use SI?

$$\begin{aligned}\text{Ans: Simple Interest} &= \frac{P \times R \times T}{100} \\ &= \frac{3000 \times 15 \times 3}{100} \\ &= \frac{135000}{100}\end{aligned}$$

So, the interest is Rs.1350

$$\begin{aligned}\text{Total amount to be returned} &= P + \text{Interest} \\ &= 3000 + 1350 \\ &= 4350\end{aligned}$$

∴ The person needs to return Rs.1350 as interest and a total of Rs.4350 to his friend.

3. In what time a sum of money will double itself at a rate of simple interest of 8% per annum?



To find the time it takes for a sum of money to double itself at a simple interest rate of 8% per annum, we can use the simple interest formula:

$$\text{Simple Interest (SI)} = \frac{\text{Principal (P)} \times \text{Rate (R)} \times \text{Time (T)}}{100}$$

When the sum doubles, the Simple Interest (SI) is equal to the Principal (P). So, we have:

$$P = \frac{P \times 8 \times T}{100}$$

Simplifying this equation:

$$100 = 8T$$

$$T = \frac{100}{8} = 12.5 \text{ years}$$

So, it will take **12.5 years** for the sum to double at an 8% simple interest rate per annum.

4. What principal will amount to Rs.570 at 4% per annum in 5 years using SI?

Ans: $A = P + \frac{P \times R \times T}{100}$

where:

$A = 570$ (Amount after interest)

$R = 4\%$

$T = 5$ years

Now, substitute the values

$$570 = P + \frac{P \times 4 \times 5}{100}$$

$$570 = P + \frac{20P}{100}$$

$$570 = P + 0.2P$$

$$570 = 1.2P$$

$$P = \frac{570}{1.2}$$

$$P = 475$$

$\therefore$ The principal is Rs. 475

Compound Interest Formula

The formula for **Compound Interest (CI)** is:

$$A = P \left(1 + \frac{R}{100} \right)^T$$

Where:

- A = Amount (Principal + Interest)
- P = Principal (Initial sum of money)
- R = Rate of interest per annum
- T = Time in years

Compound Interest Sums

1. Vijay deposited Rs. 8000 in a bank that pays 12% interest per annum compounded quarterly. Find the amount he will receive after 6 months.

$$A = P \left(1 + \frac{x}{n} \right)^{nt}$$

where:

A is the amount after interest

P is the principal

x is the annual interest rate

n is 4 (because quarter of a year is 4)

t is 0.5 years

Now, substitute the values:

$$A = 8000 \left(1 + \frac{0.12}{4} \right)^{4 \times 0.5}$$

$$A = 8000 (1 + 0.03)^2$$

$$A = 8000 \times (1.03)^2$$

$$A = 8000 \times 1.0609$$

$$A = 8487.20$$

∴ Vijay will receive Rs. 8487.20 after 6 months.

2. What will be the principal if the interest is compounded at the rate of 10% per annum for 2 years is Rs. 420.

$$A = P \left(1 + \frac{X}{100} \right)^t$$

where :

A is total amount after interest

P is principal (which we need to find)

X is annual interest rate (10%)

t is 2 years

C.I is given as Rs. 420

We know that,

$$C.I = A - P = 420$$

$$\text{Thus, } A = P + 420$$

Now, substitute the values

$$P + 420 = P \left(1 + \frac{10}{100} \right)^2$$

$$P + 420 = P (1.1)^2$$

$$P + 420 = P \times 1.21$$

$$P \times 1.21 - P = 420$$

$$P (1.21 - 1) = 420$$

$$P \times 0.21 = 420$$

$$P = \frac{420}{0.21}$$

$$0.21$$

$$P = 2000$$

∴ The principal is 2000.

3. The difference between CI and SI on a certain sum of money at 20% per annum for 3 years is Rs. 48. What is the sum of money?

Ans: When the difference between C.I and S.I is given, we can use the following formula:

$$\text{Difference} = \frac{P \times r^2 \times t}{100^2}$$

where:

P is the principal amount

r is 20%

t is 3 years

Difference is Rs. 48

Now, substitute the values:

$$48 = \frac{P \times 20^2 \times 3}{100^2}$$

$$48 = \frac{P \times 400 \times 3}{10000}$$

$$48 = \frac{1200P}{10000}$$

$$48 \times 10000 = 1200P$$

$$480000 = 1200P$$

$$P = \frac{480000}{1200}$$

$$P = 400$$

∴ Principal is Rs. 400

4. What will be the difference between CI (compounded annually) and SI at 5% per annum for 2 yrs on the amount of Rs. 2000?

Ans: Step 1: Calculate Simple Interest:

$$\text{Simple Interest (S.I)} = \text{Rs. } 200$$

Step 2: Calculate Compound Interest:

$$\text{Compound Interest (C.I)} = \text{Rs. } 205$$

Step 3: Calculate the difference:

$$\begin{aligned}\text{Difference} &= \text{C.I} - \text{S.I} \\ &= 205 - 200 \\ &= 5\end{aligned}$$

∴ The difference is of Rs. 5.

Note: Just for simplicity and for ease of understanding, I have directly written the answers of S.I and C.I. In exam you have to solve everything using formulas.

Unit 4

What is proportion?

In mathematics, proportion is a relationship that shows two ratios are equal. For example, if $a/b = c/d$ then a is to b as c is to d and we say these two ratios are in proportion.

Types of Proportion

1. Direct Proportion: When one quantity increases, the other also increases in the same ratio (e.g., doubling one quantity doubles the other).

2. Inverse Proportion: When one quantity increases, the other decreases in the same ratio (e.g., doubling one quantity halves the other).

Ratio and proportion formulas respectively



The basic formula for a ratio is:

$$\text{Ratio} = \frac{\text{Quantity 1}}{\text{Quantity 2}}$$

To represent a ratio between two quantities A and B , we write it as:

$$A : B$$

$$\frac{a}{b} = \frac{c}{d}$$

where:

- a and b are the terms of the first ratio,
- c and d are the terms of the second ratio.

To solve for an unknown variable in a proportion, you can use cross-multiplication:

$$a \cdot d = b \cdot c$$

This allows you to find the value of the unknown variable if you have the values of the other three terms.

Sums on ratio and proportion

1. Find the fourth proportional to the numbers 2, 5, 4.

Ans: Let's assume the 4th number to be x

Since they are in proportion we can use the cross multiplication method.

$$\frac{2}{5} = \frac{4}{x}$$

$$2 \times x = 5 \times 4$$

$$2x = 20$$

$$x = \frac{20}{2}$$

$$\therefore x = 10$$

$\therefore$ The fourth proportional number is 10.

2. If $A:B = 2:3$ and $B:C = 4:5$ and $C:D = 6:7$, find $A:D$.

Ans: $A:B = 2:3$
 $B:C = 4:5$

1. B is not consistent in both of these terms. So we have to make it consistent.

The least common multiple of 3 and 4 is 12, so we multiply to adjust ratios.

$$A:B = 2:3 = 2 \times 4 : 3 \times 4 = 8:12$$

$$B:C = 4:5 = 4 \times 3 : 5 \times 3 = 12:15$$

Now we have $A:B:C = 8:12:15$

2. $C:D = 6:7$ and $A:B:C = 8:12:15$
C is not consistent so we again multiply with a common multiple.

The least common multiple of 15 and 6 is 30.

$$A:B:C = 8:12:15 = 8 \times 2 : 12 \times 2 : 15 \times 2 = 16:24:30$$

$$C:D = 6:7 = 6 \times 5 : 7 \times 5 = 30:35$$

Now we have $A:B:C:D = 16:24:30:35$

$\therefore A:D = 16:35$

3. The ratio of present ages of Suresh and Mahesh is 7:5. If after six years their ages will be in the ratio of 4:3, the present age of Mahesh is?

Ans: Let present ages of Suresh and Mahesh be $7x$ and $5x$ respectively.

After six years:

Suresh's age will be $7x + 6$
Mahesh age will be $5x + 6$

According to the problem, after six years their ages will be in ratio of 4:3

$$\frac{7x+6}{5x+6} = \frac{4}{3}$$

Now, we cross-multiply:

$$\begin{aligned} 3(7x+6) &= 4(5x+6) \\ 21x+18 &= 20x+24 \\ 21x-20x &= 24-18 \\ x &= 6 \end{aligned}$$

Now we substitute $x=6$ to find present age of Mahesh.

$$5x = 5 \times 6 = 30$$

$\therefore$ Present age of Mahesh is 30

4. Anu and Renu are partners in a business. Anu invests Rs 5000 for complete year & Renu invests Rs 2500 for 6 months. What is Renu's share if they earn Rs 240 as profit?

1. Anu's investment:

Amount: Rs. 5000

Time: 12 months

Total investment value: $5000 \times 12 = 60000$

2. Renu's investment:

Amount: Rs. 2500

Time: 6 months

Investment value: $2500 \times 6 = 15000$

The ratio of their investment is:

$$60000 : 15000 = 4:1$$

Their profit will also be in 4:1 ratio.

$$\text{Renu's share} = \frac{1}{4+1} \times 240 = \frac{1}{5} \times 240 = 48$$

$\therefore$ Renu's share of profit is Rs. 48

5. A, B and C signed up a partnership. A invested Rs. 2000/-. B invested Rs. 4000/- and C invested Rs. 3000/-. At the end of 5 months A withdrawn Rs. 400/- of his capital and at the end of 11 months B increased his capital by Rs. 1800/-. First year profit is Rs 1605. What is B's share of total profit?

1. A's investment:

- Initial investment = Rs. 2000
- After 5 months, A withdrew Rs. 400, so new investment = Rs. 1600

A's investment calculation:

- Rs. 2000 for first 5 months
 $= 2000 \times 5 = 10000$
- Rs. 1600 for remaining 7 months
 $= 1600 \times 7 = 11200$

$$\therefore \text{Total investment by A} = 10000 + 11200 \\ = 21200$$

2. B's investment:

- Initial investment: Rs. 4000
- End of 11 months, B increased his capital by 1800, so it becomes Rs. 5800.

B's investment calculation:

- Rs. 4000 for first 11 months:
 $4000 \times 11 = 44000$
- Rs. 5800 for last month:
 $5800 \times 1 = 5800$

$$\therefore \text{Total investment by B} = 44000 + 5800 \\ = 49800$$

3. C's investment:

- Initial investment: Rs. 3000

C's investment calculation:

- $\text{Rs. } 3000 \times 12 \text{ months}$
 $= 36000$

∴ Total investment by C = 36000

Total investment ratios are:

$$\begin{aligned} A:B:C &= 21200:49800:36000 \\ &= 212:498:360 \text{ (dividing by 100)} \end{aligned}$$

B's profit will be in proportion to his investment

$$\text{B's share} = \frac{498}{212+498+360} \times 1605 \text{ (Total profit)}$$

$$= \frac{498}{1070} \times 1605$$

$$= 747$$

∴ B's share of total profit is Rs. 747.

Sums on mixtures and alligations

1. 3. A milkman mixes 1 liter of water to x liters of milk and sells the mixture for Rs. 20/liter. If the price of pure milk is Rs 25/liter then find the value of x .

Ans: Given:

Price of water = Rs. 0/litre (free)
Price of milk = Rs. 25/litre
Price of mixture = Rs. 20/litre
Water = 1 litre
Milk = x litre

Solution:

Difference between prices:

1. Milk price and Mixture price = $25 - 20 = 5$
2. Water price and Mixture price = $0 - 20 = 20$

The ratio of water to milk is:

$$5 : 20 = 1 : 4$$

Now, from the ratio we can conclude that for every 1 litre of water, there are 4 litres of milk.

$\therefore x = 4$

2. 2 liters of water is added to 8 liters of 25% solution of alcohol in water. What is the strength of alcohol now?

Given:

1. Initial Solution: 8 l of 25% alcohol solution.
2. Alcohol in solution: $25\% \times 8 = \frac{25}{100} \times 8 = 2 \text{ l}$
3. Water in solution: $8 - 2 = 6 \text{ l}$
4. Additional water added: 2 l

Total solution after adding water:

$$8 + 2 = 10 \text{ l}$$

Strength of Alcohol = $\frac{\text{Alcohol}}{\text{Total solution}} \times 100$

$$= \frac{2}{10} \times 100$$
$$= 20$$

∴ Strength of Alcohol is 20%

Note: I did not find any sums on alligations from pyq's so i added only 2 because they are still there in the syllabus. All the alligation sums are based on percentages and ratios so understand the logic of the question and it will be easy.

Unit 5

Mean Formula and example

$$m = \frac{\text{sum of the terms}}{\text{number of terms}}$$

m = mean

Finding the Mean

Here are the marks out of 25 scored by 5 students in a Maths test.



To work out the average (Mean) you add up all the scores and divide by the total number of student.

$$\text{Mean} = \frac{\text{Total marks}}{\text{No of students}}$$

What is the mean of this dataset?

$$\text{Mean} = \frac{85}{5} = 17$$

Mean

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Mode(Grouped data) formula and example

The formula for calculating the mode of grouped data is:

$$Mode = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) h$$

Here,

- L : The lower limit of the modal class
- h : The size of the class interval
- f_1 : The frequency of the modal class
- f_0 : The frequency of the class preceding the modal class
- f_2 : The frequency of the class succeeding the modal class

Here the maximum frequency is 14 which is in 130-135 class interval.

Modal class = 130-135

$L = 130, h = 5, f_1 = 14, f_2 = 10, f_0 = 7$

$\Rightarrow Mode = 130 + ((14-7)/(2 \times 14 - 7 - 10)) \times 5$

$\Rightarrow Mode = 130 + 3.18 = 133.18$

So, modal height = 133.18

Mode (ungrouped data) formula and example

The mode for ungrouped data is the value that appears most frequently in the data set: 

1. Arrange the data in ascending or descending order 
2. Identify the repeated values and count their frequency 
3. The value with the highest frequency is the mode 

For example, in the data set 6, 8, 9, 3, 4, 6, 7, 6, 3, the value 6 appears the most, so the mode is 6. 

Median Formula

if n is odd,

$$\text{median} = \left(\frac{n+1}{2} \right)^{th}$$

if n is even,

$$\text{median} = \frac{\left(\frac{n}{2} \right)^{th} + \left(\frac{n}{2} + 1 \right)^{th}}{2}$$

n = number of terms

th = $n(th)$ number

Q. Find Median of the following dataset:

8, 6, 12, 15, 18

Ans: Step 1: Arrange data in ascending order.

5, 8, 12, 15, 18

Step 2: find even or odd?

There are 5 numbers which means an odd dataset.

Step 3: Apply formula and solve

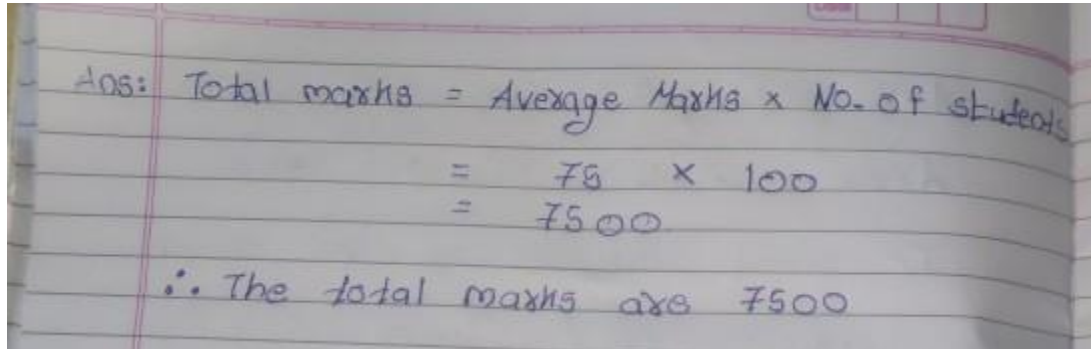
$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} = \left(\frac{5+1}{2} \right) = 3^{\text{th}}$$

The 3rd number in the dataset is 12.

$\therefore$ Median = 12

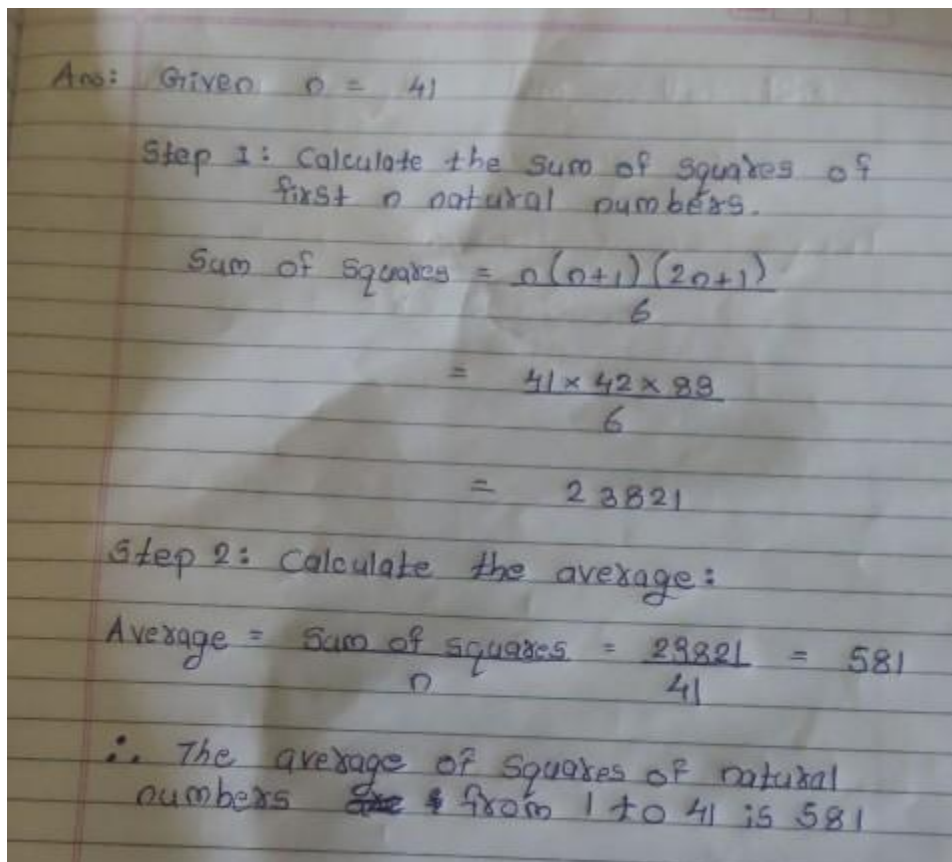
Sums on Averages and Ages

1. The average marks obtained by 100 students are 75. Find the total marks.



Ans: Total marks = Average Marks $\times$ No. of students
 $= 75 \times 100$
 $= 7500$
 $\therefore$ The total marks are 7500

2. Find the average of squares of the natural numbers from 1 to 41.



Ans: Given $n = 41$

Step 1: Calculate the sum of squares of first n natural numbers.

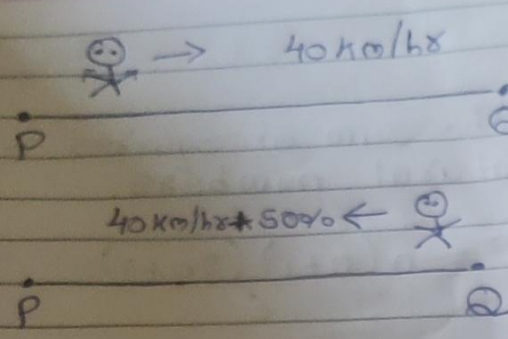
$$\text{Sum of Squares} = \frac{n(n+1)(2n+1)}{6}$$
$$= \frac{41 \times 42 \times 83}{6}$$
$$= 28821$$

Step 2: Calculate the average:

$$\text{Average} = \frac{\text{Sum of squares}}{n} = \frac{28821}{41} = 581$$

$\therefore$ The average of squares of natural numbers ~~are~~ from 1 to 41 is 581

3. A person travels from P to Q at a speed of 40 km/hr and returns by increasing his speed by 50%. Find his average speed for both the trips.



Ans: Given:

Speed from P to Q (s_1) = 40 km/hr
Speed from Q to P (s_2) = $40 + 50\% = 60$ km/hr

Since, the distances are equal we can use the following formulas:

$$\text{Average speed} = \frac{2 \times \text{speed}_1 \times \text{speed}_2}{\text{speed}_1 + \text{speed}_2}$$
$$= \frac{2 \times 40 \times 60}{40 + 60}$$
$$= 48 \text{ km/hr}$$

$\therefore$ The average speed for trip is 48 km/hr.

4. The average age of a husband, his wife and son 3 years ago was 27 years and that of his wife and son 5 years ago was 20 years. What is the husband's present age?

Ans: let H be Husband's present age
W be Wife's present age
S be Son's present age

1. Average age of Husband, wife and son 3 years ago was 27 years.

$$\frac{(H-3) + (W-3) + (S-3)}{3} = 27$$

Multiply both sides by 3:

$$(H-3) + (W-3) + (S-3) = 81$$
$$= H + W + S - 9 = 81$$
$$= H + W + S = 90 \quad \text{(equation (1))}$$

2. Average of wife and son 5 years ago was 20 years.

$$\frac{(W-5) + (S-5)}{2} = 20$$

Multiply both sides by 2:

$$(W-5) + (S-5) = 40$$
$$W + S - 10 = 40$$
$$W + S = 50 \quad \text{(equation (2))}$$

3. Subtract equⁿ (2) from equⁿ (1)

$$(H + W + S) - (W + S) = 90 - 50$$
$$= 40$$

∴ Present age of husband is 40

5. A is two years older than B who is twice as old as C. If the total of the ages of A, B and C be 27, then how old is B?

Ans: Let the age of C be C.

Since B is twice as old as C, B's age will be $2C$.

A is two years older than B, so A's age will be $2C + 2$.

We are given:

$$A + B + C = 27$$

Substitute the expressions for A, B, C

$$(2C + 2) + 2C + C = 27$$

$$= 2C + 2 + 2C + C = 27$$

$$= 5C + 2 = 27$$

$$= 5C = 27 - 2$$

$$5C = 25$$

$$C = \frac{25}{5}$$

$$C = 5$$

Find B's age by substituting value of C

Since $B = 2C$ and $C = 5$

$$B = 2 \times 5$$

$$= 10$$

$\therefore$ B is 10 years old

Note: I did not find any sums related to mean median and mode in PYQ's. But since they are included in our syllabus, just have a glance over the formulas to be prepared for any unexpected surprises in exams.

Unit 6

Sums on Time and Work

1. A pipe can fill a tank in 8 hours and another can empty it in 16 hours. If both the pipes are opened simultaneously, find the time taken to fill the tank.

Ans: Filling pipe rate = $\frac{1}{8}$ of the tank per hour

Emptying pipe rate = $-\frac{1}{16}$ of tank per hour

$$\text{Net rate} = \frac{1}{8} - \frac{1}{16}$$

We have to make denominators common to perform subtraction.

$$\frac{1}{8} \times 2 = \frac{2}{16}$$

$$-\frac{1}{16} \times 1 = -\frac{1}{16}$$

$$\therefore \text{Net rate} = \frac{2}{16} - \frac{1}{16} = \frac{1}{16} = 16$$

$\therefore$ The tank can be filled in 16 hours when both pipes are opened.

2. Two pipes P and Q can fill a tank separately in 20 minutes and 30 minutes. If both the pipes are opened simultaneously, find the time taken to fill the tank.

Ans: Pipe P's rate = $\frac{1}{20}$ of tank per minute

Pipe Q's rate = $\frac{1}{30}$ of tank per minute

Combined rate = $\frac{1}{20} + \frac{1}{30}$

To perform addition, we have to make the denominators common for both of them.

$$\frac{1}{20} \times 3 = \frac{3}{60}$$

$$\frac{1}{30} \times 2 = \frac{2}{60}$$

$$\text{Combined rate} = \frac{3}{60} + \frac{2}{60} = \frac{5}{60}$$

1 hour has 60 minutes, so we will divide 60 by 5.

$$\frac{60}{5} = 12$$

∴ The tank will be filled in 12 minutes when both pipes open simultaneously.

3. Two pipes can fill a tank in 8 hours and 12 hours respectively, while a third pipe can empty it in 6 hours. If all the pipes are opened simultaneously, find the time taken to fill the tank.

$$\text{Ans: Rate of first pipe} = \frac{1}{8}$$

$$\text{Rate of second pipe} = \frac{1}{12}$$

$$\text{Rate of third pipe} = -\frac{1}{6}$$

$$\text{Net rate} = \frac{1}{8} + \frac{1}{12} - \frac{1}{6}$$

To perform and solve this equation, we have to find least common denominator for every fraction.

$$\frac{1}{8} \times 3 = \frac{3}{24}$$

$$\frac{1}{12} \times 2 = \frac{2}{24}$$

$$-\frac{1}{6} \times 4 = -\frac{4}{24}$$

$$\therefore \text{Net rate} = \frac{3}{24} + \frac{2}{24} - \frac{4}{24}$$

$$= \frac{1}{24}$$

$$= 24$$

$\therefore$ The tank will be filled in 24 hours.

4. Three persons can finish a piece of work in 10, 15 and 18 days respectively. If all of them work together, what is the time taken to finish the work?

$$\text{Person A's rate} = \frac{1}{10} \text{ of work per day}$$

$$\text{Person B's rate} = \frac{1}{15} \text{ of work per day}$$

$$\text{Person C's rate} = \frac{1}{18} \text{ of work per day}$$

$$\text{Combined rate} = \frac{1}{10} + \frac{1}{15} + \frac{1}{18}$$

To add the fractions, we have to find the least common denominator.

$$\frac{1}{10} \times 9 = \frac{9}{90}$$

$$\frac{1}{15} \times 6 = \frac{6}{90}$$

$$\frac{1}{18} \times 5 = \frac{5}{90}$$

$$\therefore \text{Combined rate} = \frac{9}{90} + \frac{6}{90} + \frac{5}{90}$$

$$= \frac{20}{90}$$

$$= \frac{2}{9}$$

Reverse the fraction to have bigger number on top

$$\frac{9}{2} = 4.5$$

$\therefore$ The work will be completed in 4.5 days

5. Sohan takes twice as much time as Mohan to complete a work and Rohan does it at the same time as Sohan and Mohan together. If all of them working together can finish the work in 6 days, then find the time taken by each of them separately to finish the work.

Ans: 1. Let Mohan take x days to finish work

2. Sohan takes twice as much time as Mohan, so Sohan will take $2x$ days

3. Rohan can complete the work in same time as Sohan and Mohan working together, which can be calculated as:

$$\frac{x \times 2x}{x + 2x} = \frac{2x}{3} \text{ days}$$

Now we calculate their individual
Now, we calculate their combined rate:

$$\frac{1}{x} + \frac{1}{2x} + \frac{3}{2x} = \frac{6}{2x} = \frac{3}{x}$$

It is given that they take 6 days to finish the work together, so:

$$\frac{1}{6} = \frac{3}{x}$$

Now we solve for x by cross multiplying

$$\begin{aligned} 6 \times 3 &= 1 \times x = x = 18 & 6 \times 3 &= 18 \\ & 1 \times x &= x \\ \therefore x &= 18 \end{aligned}$$

Now, we calculate time for individual person.

$$\text{Mohan} = x = 18$$

$$\text{Sohan} = 2x = 36$$

$$\text{Rohan} = \frac{2x}{3} = \frac{2 \times 18}{3} = 12$$

∴ Mohan takes 18 days, Sohan 36 days and Rohan 12 days.

6. P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If both P and Q work together working 8 hours a day, in how many days can they complete the work?

Ans: Step 1: Calculate the total working hours

1. P takes 12 days working 8 hours a day,
So total work for P = $12 \times 8 = 96$ hours

2. Q takes 8 days working 10 hours a day,
So total work for Q = $8 \times 10 = 80$ hours.

P's rate of work per hour = $\frac{1}{96}$

Q's rate of work per hour = $\frac{1}{80}$

Combined work rate = $\frac{1}{96} + \frac{1}{80}$

We have to make the denominators common to perform addition

$$\frac{1}{96} \times 5 = \frac{5}{480}$$

$$\frac{1}{80} \times 6 = \frac{6}{480}$$

$$\therefore \text{Combined rate} = \frac{5}{480} + \frac{6}{480} = \frac{11}{480}$$

When both of them work 8 hours a day, their combined work rate will be:

$$8 \times \frac{11}{480} = \frac{88}{480} = \frac{11}{60}$$

For substitution, we will adjust the fraction to have the bigger number on the top

$$\frac{60}{11} = 5.45$$

$\therefore$ P and Q will complete the work together in 5.45 days.

Note:

1. Whenever the questions are about pipes, look for the keywords "fill" and "empty ". When the pipe is said to fill the tank, it will be a positive fraction, when it is said empty, it will be a negative fraction.

2. This whole chapter sums can be solved using the same pattern. Which is:

1. Write the people/pipes work rate in fraction.
 2. Try to add them, if denominator is same.
 3. If denominator is not same then make them same by multiplying with the L.C.D
 4. When the denominators are same add the Fractions together.
 5. Substitution the Fractions if the can be i.e if $\frac{5}{20}$ is your fraction you can substitute it further to $\frac{1}{4}$.
 6. At last reverse the fraction to have the bigger number as numerator and smaller number as denominator.
 7. Divide the number and get your final answer.
-

Unit 7

Important Formulas

1. **Distance** = Speed x Time

2. **Time** = Distance / Speed

3. **Speed** = Distance/ Time

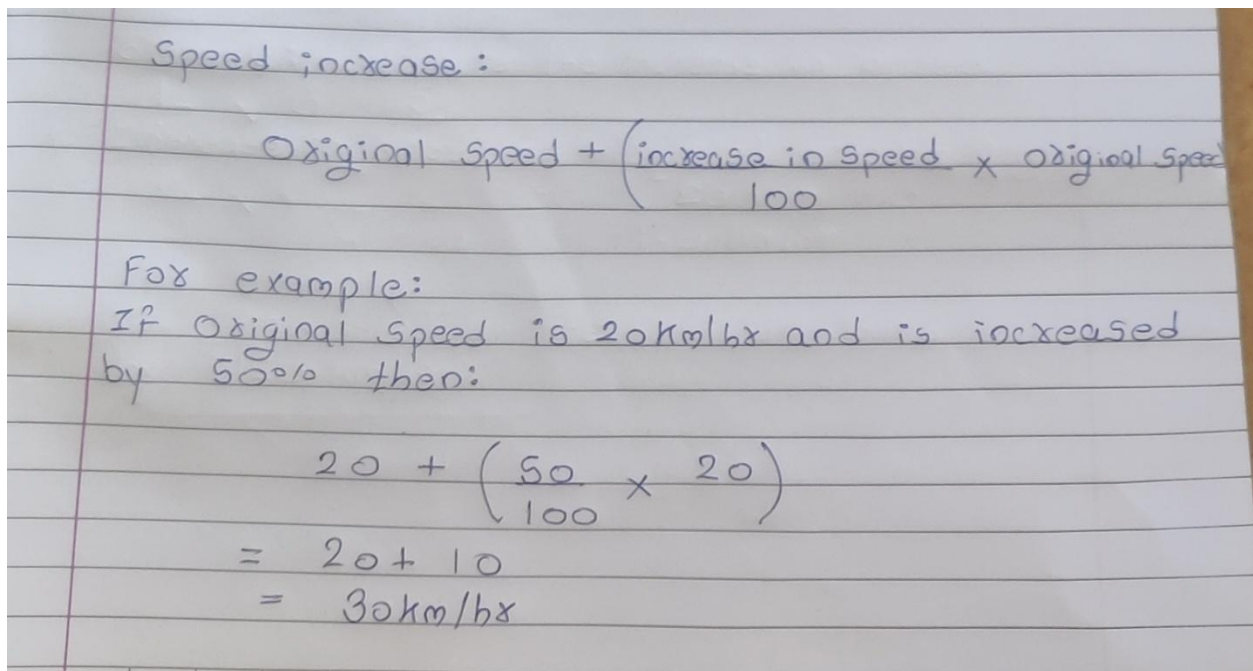
4. **Average speed**=

$$2 \times \text{speed}^1 \times \text{speed}^2 / \text{speed}^1 + \text{speed}^2$$

4. To **convert** a speed given in **m/s** into a speed in **km/h**, multiply with 18/5

5. To **convert** a speed given in **km/h** into a speed in **m/s**, multiply with 5/18

6. **Speed increase:**



Speed increase:

$$\text{Original Speed} + \left(\frac{\text{increase in speed}}{100} \times \text{original speed} \right)$$

For example:

If original speed is 20 km/hr and is increased by 50% then:

$$20 + \left(\frac{50}{100} \times 20 \right)$$
$$= 20 + 10$$
$$= 30 \text{ km/hr}$$

Sums on Time, Speed and Distance

1. A car covers its journey at the speed of 80km/hr in 10hours. If the same distance is to be covered in 4 hours, by how much the speed of car will have to increase?

Ans: Step 1: Calculate the total distance

The car covers the journey at a speed of 80 km/hr in 10 hours.

$$\begin{aligned}\text{Distance} &= \text{Speed} \times \text{Time} \\ &= 80 \times 10 \\ &= 800 \text{ km}\end{aligned}$$

Step 2: Calculate the required speed to cover the distance in 4 hours

$$\begin{aligned}\text{Required speed} &= \frac{\text{Distance}}{\text{Time}} \\ &= \frac{800}{4} \\ &= 200 \text{ km/hr}\end{aligned}$$

Step 3: Calculate the increase in Speed

$$\begin{aligned}\text{Increase in speed} &= \text{Required speed} - \text{Original speed} \\ &= 200 - 80 \\ &= 120 \text{ km/hr}\end{aligned}$$

∴ The speed of the car will have to increase by 120 km/hr.

2. Two persons are running in the same direction with the speeds of 3 km/hr and 6 km/hr. A truck comes running from behind and passes them in 9 and 10 seconds respectively. What is the speed of the truck?

Ans: Given:

Speed of person 1: 3 km/hr

Speed of person 2: 6 km/hr

Since, it took seconds for the truck to pass, it is better to convert km/hr into m/s.

Person 1: $3 \text{ km/hr} \times 0.27 = 0.81 \text{ m/s}$

Person 2: $6 \text{ km/hr} \times 0.27 = 1.62 \text{ m/s}$

Let the truck's speed be $v \text{ m/s}$

1. Relative speed with respect to person 1:

$$v - 0.81$$

2. Relative speed with respect to person 2:

$$v - 1.62$$

The distance of the truck is not given, so we have to calculate that first:

Person 1: $d = (v - 0.81) \times 9$ $\left(\because \text{Distance} = S \times T \right)$

Person 2: $d = (v - 1.62) \times 10$

Since, d is common in both equations we can solve them together

$$\begin{aligned}
 (v - 0.81) \times 9 &= (v - 1.62) \times 10 \\
 = 9v - 7.29 &= 10v - 16.2 \\
 9v - 10v &= -16.2 + 7.29 \\
 -1v &= -8.91 \\
 v &= \frac{-8.91}{-1} \\
 v &= 8.91 \text{ km/hr m/s}
 \end{aligned}$$

Now we convert our speed of the truck to km/hr

$$8.91 \text{ m/s} \times 3.6 = 32.0 \text{ km/hr}$$

3. A car travels from P to Q with 15 km/hr and returns with 30 km/hr. What is the average speed of journey?

Ans: Given:

Speed from:

$$P \rightarrow Q = 15 \text{ km/hr} \quad \text{--- (S1)}$$

$$Q \rightarrow P = 30 \text{ km/hr} \quad \text{--- (S2)}$$

$$\text{Average speed} = \frac{2 \times \text{Speed}_1 \times \text{Speed}_2}{\text{Speed}_1 + \text{Speed}_2}$$

$$= \frac{2 \times 15 \times 30}{15 + 30}$$

$$= 20$$

$\therefore$ The average speed of the car is 20 km/hr

4. A biker goes from his starting point to his destination with 20 km/hr, but returns with his speed increased by 50%. What is the average speed of his journey?

Given:

Speed 1 = 20 km/hr

Speed 2 = 50% of 20 km/hr increase = 30 km/hr

$$\text{Average speed} = \frac{2 \times \text{Speed 1} \times \text{Speed 2}}{\text{Speed 1} + \text{Speed 2}}$$

$$= \frac{2 \times 20 \times 30}{20 + 30}$$

$$= 24$$

∴ Average speed of biker is 24 km/hr

Important formulas on Trains

Trains

1. Time to cross a Pole or stationary object:

$$\text{Time} = \frac{\text{length of train}}{\text{Speed of train (m/s)}}$$

2. Time to cross a platform:

$$\text{Time} = \frac{\text{length of train} + \text{length of platform}}{\text{Speed of train (m/s)}}$$

3. Relative speed when moving in same direction:

$$\text{Relative Speed} = \text{Speed of Train 1} - \text{Speed of Train 2}$$

4. Relative speed when moving in opposite direction:

$$\text{Relative speed} = \text{Speed of Train 1} + \text{Speed of Train 2}$$

5. Time to cross another train in opposite direction:

$$\text{Time} = \frac{\text{length of Train 1} + \text{length of Train 2}}{\text{Relative Speed}}$$

6. length of train = Speed $\times$ time

Sums on Trains

1. A train running at a speed of 60km/hr crosses a pole in 9 seconds, what is the length of the train?

Ans: Given:

Speed of train = 60km/hr

Time to cross the pole = 9 seconds

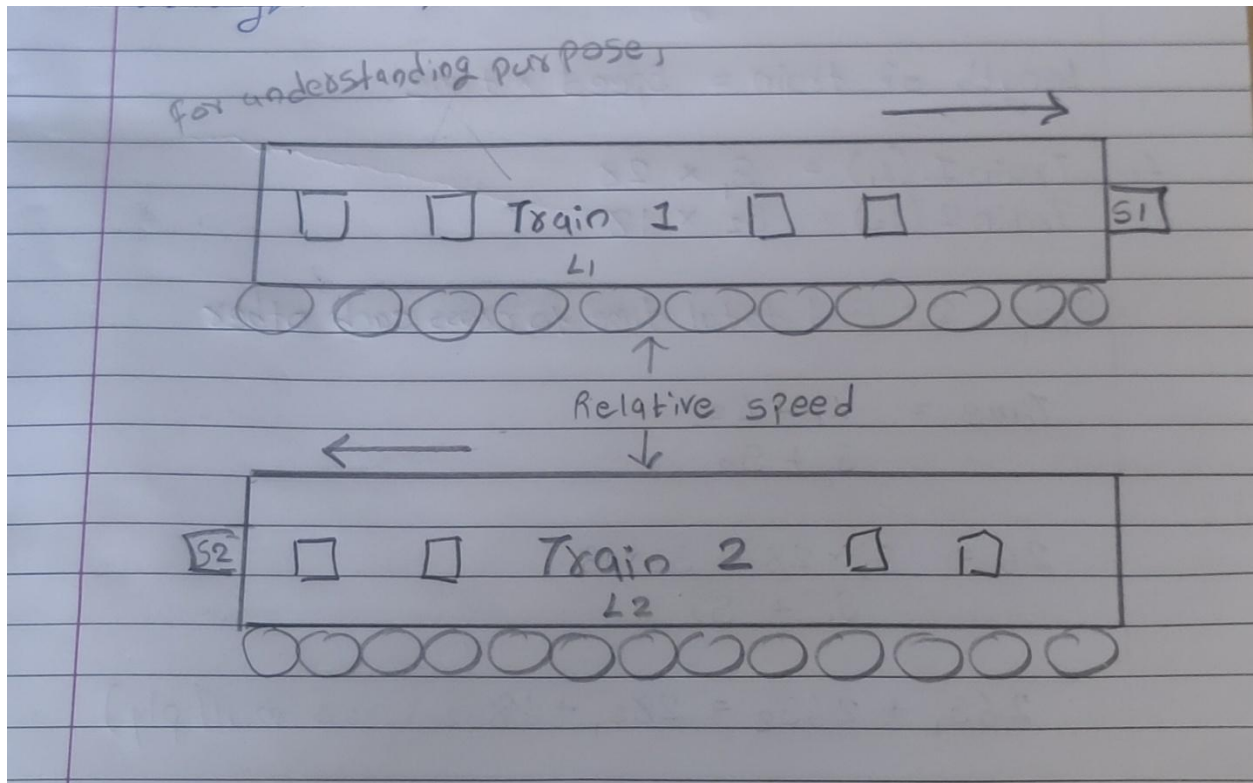
We will first convert km/hr in m/s.

$$60 \times \frac{5}{18} = 16.67 \text{ m/s}$$

$$\begin{aligned} \text{length of train} &= \text{Speed} \times \text{time} \\ &= 16.67 \times 9 \\ &= 150 \end{aligned}$$

$\therefore$ length of train = 150 meters

2. Time is taken by two trains running in opposite directions to cross a man standing on the platform in 28 seconds and 18 seconds respectively. It took 26 seconds for the trains to cross each other. What is the ratio of their speeds?



Ans: When two trains cross each other while travelling in opposite directions, the total distance covered is sum of their lengths.

The relative speed is also the sum of their speeds.

Let:

L_1 = Length of train 1

L_2 = Length of train 2

S_1 = Speed of train 1

S_2 = Speed of train 2

Step 1: Find length of each train

Length of train = Speed $\times$ time

1. Train 1 (L_1) = $S_1 \times 28$

2. Train 2 (L_2) = $S_2 \times 18$

Step 2: Find total time to cross each other

$$\text{Time} = \frac{L_1 + L_2}{S_1 + S_2}$$

$$26 = \frac{S_1 \times 28 + S_2 \times 18}{S_1 + S_2}$$

$$26S_1 + 26S_2 = 28S_1 + 18S_2 \text{ (cross multiply)}$$

Rearranging the equation for substitution:

$$28s_1 - 26s_1 = 26s_2 - 18s_2$$

$$2s_1 = 8s_2$$

$$\frac{s_1}{s_2} = \frac{8}{2}$$

$$= \frac{4}{1}$$

$\therefore$ The ratio of their speed is 4:1

3. A train 145 meters long is running with a speed of 25 km/hr. In what time will it pass a man who is walking at 4 km/hr in the direction opposite to that of the train?

Ans: Step 1: Calculate the relative Speed

$$\begin{aligned}\text{Relative Speed} &= \text{Speed of train} + \text{Speed of man} \\ &= 25 + 4 \\ &= 29 \text{ km/hr}\end{aligned}$$

Step 2: Convert relative speed to m/s

$$29 \text{ km/hr} \times \frac{5}{18} = 8.06 \text{ m/s}$$

Step 3: Calculate time:

$$\begin{aligned}\text{Time} &= \frac{\text{Length of train}}{\text{Relative speed}} = \frac{145}{8.06} \\ &= 18\end{aligned}$$

∴ The train will pass the man in 18 seconds

4. A train 300 m long is running at a speed of 54 km/hr. In what time will it pass a bridge 150 m long

Ans: Given:

Length of train = 300m

Length of bridge = 150m

Speed of train = 54 km/hr

Step 1: Convert Speed to m/s.

$$\text{Speed} = 54 \text{ km/hr} \times \frac{5}{18} = 15 \text{ m/s}$$

Step 2: Calculate total Distance

$$\begin{aligned} \text{Total Distance} &= \text{Length of train} + \text{Length of Bridge} \\ &= 300 + 150 \\ &= 450 \text{ m} \end{aligned}$$

Step 3: Calculate the time

$$\text{Time} = \frac{\text{Total Distance}}{\text{Speed}} = \frac{450}{15} = 30$$

∴ The train will pass the bridge in 30 seconds.

5. Two trains of lengths 100 meters and 120 meters are running in the same direction with speeds 30 km/hr and 40 km/hr respectively. In what time (approx.) will they pass each other?

Ans. Given:

Length of Train 1 (L_1) = 100m

Length of Train 2 (L_2) = 120m

Speed of Train 1 (S_1) = 30 km/hr

Speed of Train 2 (S_2) = 40 km/hr

Step 1: Calculate relative speed

$$\begin{aligned}\text{Relative Speed} &= S_2 - S_1 \\ &= 40 - 30 \\ &= 10 \text{ km/hr}\end{aligned}$$

Step 2: Convert km/hr in m/s

$$10 \text{ km/hr} \times \frac{5}{18} = 2.78 \text{ m/s}$$

Step 3: Calculate total Distance

$$\begin{aligned}\text{Total Distance} &= L_1 + L_2 \\ &= 100 + 120 \\ &= 220 \text{ meters}\end{aligned}$$

Step 4: Calculate time

$$\text{Time} = \frac{\text{Total Distance}}{\text{Relative Speed}} = \frac{220}{2.78} \approx 79.14$$

∴ The trains will pass each other in 79 seconds

Formulas on Swimmers/boats and River

1. **Upward Stream** = Speed of Swimmer - Speed of river.
2. **Downward Stream** = Speed of Swimmer + Speed of river.

3. Speed in still water =

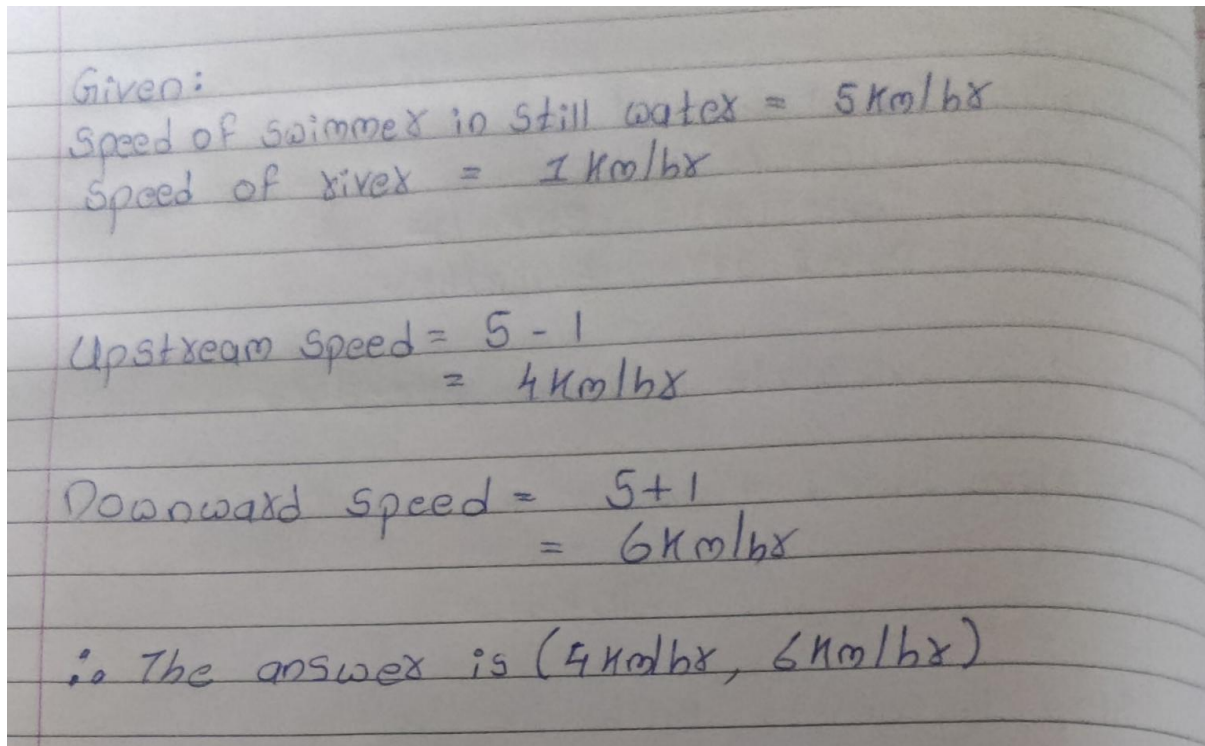
Downstream speed + upstream speed / 2

4. Speed of the river =

Downstream speed - upstream speed / 2

Sums on Swimmers

1. The speed of a swimmer in still water is 5 km/hr and the speed of the river is 1 km/hr. Find the upstream and downstream speeds of the swimmer.



2. A person can swim in water with a speed of 13km/hr in still water. If the stream of the water is 4 km/hr, what will be the time taken by the person to reach 68 km downstream?

$$\text{Ans: Downstream speed} = 13 + 4 \\ = 17 \text{ km/hr}$$

$$\text{Time taken} = \frac{\text{Distance}}{\text{Speed}}$$

$$= \frac{68}{17}$$

$$= 4$$

$\therefore$ The time taken is 4 hours.

3. A boat is rowed down a river 40 km in 5 hours and up the river 21 km in 7 hours. What is the speed of the boat and the river?

Ans: Given:

Downstream Distance = 40 km, time = 5 hours

Upstream Distance = 21 km, time = 7 hours

Step 1: Calculate downstream and upstream speeds

$$\text{Downstream Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{40}{5} = 8 \text{ km/hr}$$

$$\text{Upstream Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{21}{7} = 3 \text{ km/hr}$$

Speed of

Step 2: Calculate speed of boat in still water

$$\frac{\text{Downstream Speed} + \text{Upstream Speed}}{2}$$

$$= \frac{8 + 3}{2}$$

$$= 5.5 \text{ km/hr}$$

Step 3: Calculate speed of river:

$$\frac{\text{Downstream Speed} - \text{Upstream Speed}}{2}$$

$$= \frac{8 - 3}{2}$$

$$= 2.5 \text{ km/hr}$$

∴ Speed of boat in still water is 5.5 km/hr and speed of river is 2.5 km/hr.

Unit 8

Permutations and Combinations

1. Permutations formula:

$$\text{Permutation}$$
$${}^n P_r = \frac{n!}{(n-r)!}$$

where,

- n = Number of Total Objects
- r = Number of Objects Chosen at Once
- $0 \leq r \leq n$

2. Permutations formula for a word:

$$\text{Permutations} = \frac{n!}{p_1! \times p_2! \times p_3! \dots}$$

Here,

n = Total number of letters

p_1 = Most repeated frequency of a letter in the word

p_2 = 2nd most repeated

p_3 = 3rd most repeated

p_4 = 4th and so on....

Example:

Question Find Permutations of the word "TOMORROW"

~~Here, $n = 8$~~

~~Repeated letters~~
~~TOMORROW~~

Ans:

Here,

$$n = 8$$

$O(p_1) = 3$ times repeated

$R(p_2) = 2$ times repeated

$$\therefore \text{permutations} = \frac{n!}{p_1! \times p_2!}$$

$$= \frac{8!}{3! \times 2!} = 3360$$

Note: the Permutation formula No.1 and No.2 have different use cases. Use formula No.2 only when you encounter questions where you have to find permutations of a word like shown in the image above. For rest of the questions related to permutations use Formula No.1

3. Combinations Formula:

Formula

$${}_nC_r = \frac{n!}{r!(n-r)!}$$

${}_nC_r$ = number of combinations

n = total number of objects in the set

r = number of choosing objects from the set

4. Consonants: Every letter in English alphabet except "a,e,i,o,u"

Sums on Permutations, Combinations and Factorials

1. Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

Step 2: Select 2 vowels from 4

$${}^4C_2 = \frac{4!}{2!(4-2)!} = \frac{24}{4} = 6$$

Step 3: Arrange the selected 3 consonants and 2 vowels in a word

3 consonants + 2 vowels = 5 letters

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Step 4: Find total number of words

$$35 \times 6 \times 120 \\ = 25,200$$

$\therefore$ Total 25,200 words can be formed.

7.
Ans: step 1: select 3 consonants from 7

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$${}^7C_3 = \frac{7!}{3!(7-3)!}$$

$$= \frac{5040}{144} = 35$$

2. Find the permutations of the word "ALLAHABAD"

Ans: Here,
 $n = 9$
 $A(p_1) = 4$ times repeated
 $L(p_2) = 2$ times repeated

$$\text{Permutations} = \frac{n!}{p_1! \times p_2!}$$
$$= \frac{9!}{4! \times 2!}$$
$$= \frac{362880}{48}$$
$$= 7560$$

$\therefore$ The number of distinct permutations of the word "ALLAHABAD" is 7560.

3. There are 15 buses from Pune to Mumbai and back. In how many ways can a person go from Pune to Mumbai and back?

Ans: Buses from Mumbai to Pune = 15
Buses from Pune to Mumbai = 15

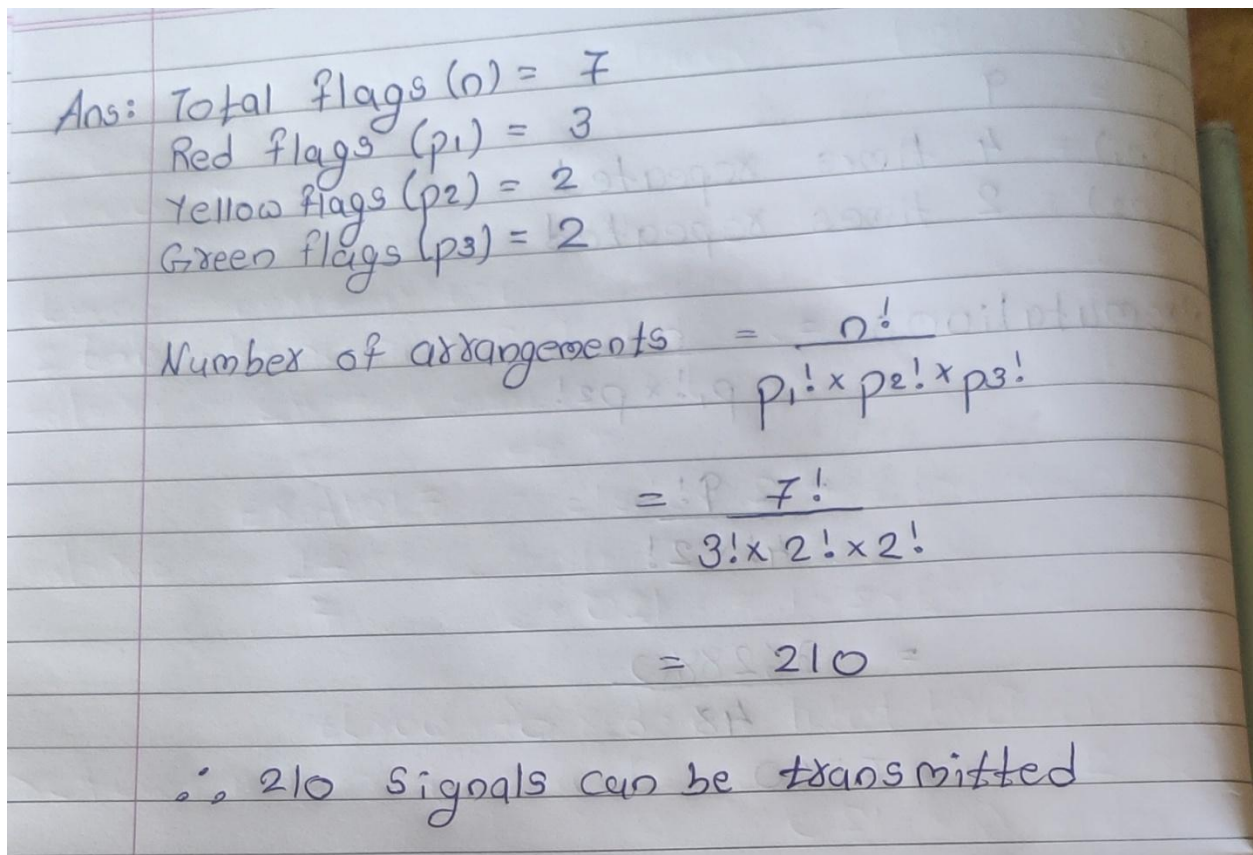
$$\text{Total number of ways to travel} = 15 \times 15$$
$$= 225$$

$\therefore$ A person can travel in 225 ways.

4. In how many ways can seven coins be put in four different boxes?

Ans: Each coin has 4 choices of boxes where it can be placed. Since the placement of one coin is independent of the others: 4^7

5. There are 3 red, 2 yellow and 2 green flags. Using all the flags at a time, how many signals can be transmitted?



Ans: Total flags $(n) = 7$
Red flags $(p_1) = 3$
Yellow flags $(p_2) = 2$
Green flags $(p_3) = 2$

$$\text{Number of arrangements} = \frac{n!}{p_1! \times p_2! \times p_3!}$$
$$= \frac{7!}{3! \times 2! \times 2!}$$
$$= 210$$

$\therefore$ 210 signals can be transmitted

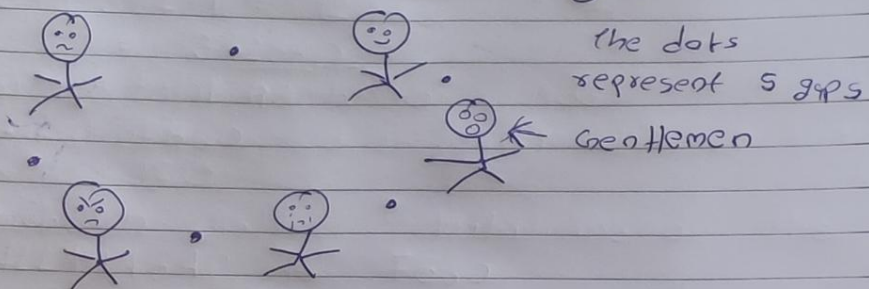
6. There are 5 gentlemen and 4 ladies standing in a circle. If no two ladies can stand next to each other, in how many ways can they stand?

Ans: Since the gentlemen are standing in a circle, their arrangement is considered circular.

The number of ways to arrange the gentlemen:

$$4! = 24$$

The number of ways to arrange ladies to ensure no ladies stand to each other, the 4 ladies must be placed in 4 of these 5 gaps



The number of ways to arrange the ladies:

$$4! = 24$$

Total number of arrangements:

Arrangements of gentlemen $\times$ Arrangement of ladies $\times$
Choice of gaps for ladies

$$24 \times 5 \times 24 = 2880$$

$\therefore$ There are total 2880 ways to stand

7. In an examination, there are two groups of questions containing 4 questions each. A student has to attempt 5 questions in all, but he cannot attempt more than 3 questions from any group. In how many ways can he attempt the questions?

Ans:

Group A	Group B
4 Questions	4 Questions

To help with visualization, not necessary to draw in exam

Attempt total 5 questions in any way
constraint is to only pick at most 3 questions from any group.

Case 1: 3 questions from Group A and 2 questions from Group B

1. Number of ways to choose 3 questions from Group A:

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{4!}{3!(4-3)!} = 4$$

2. Number of ways to choose 2 questions from Group B:

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{4!}{2!(4-2)!} = 6$$

Total for this case:

$$4 \times 6 = 24$$

Case 2: 2 questions from Group A and 3 questions from Group B

1. Number of ways to choose 2 questions from Group A:

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{4!}{2!(4-2)!} = 6$$

2. Number of ways to choose 3 questions from Group B:

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{4!}{3!(4-3)!} = 4$$

Total for this case:

$$6 \times 4 = 24$$

Adding results from both cases:

$$24 + 24 = 48$$

$\therefore$ The student can attempt the questions in 48 ways.

Note:

Key Difference Between Linear and Circular Arrangements:

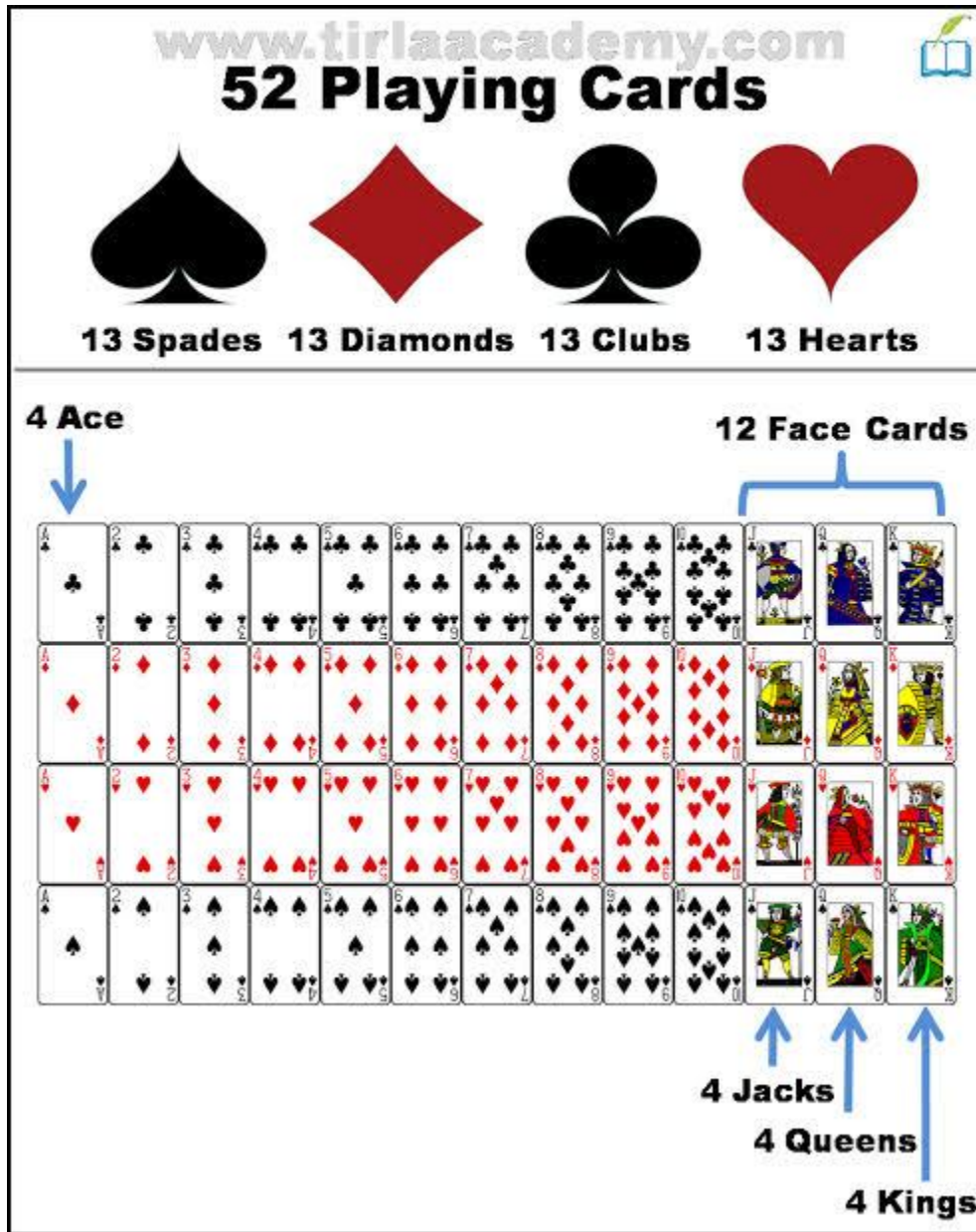
1. Linear Arrangement:

- In a linear arrangement, each position is unique, and all arrangements are distinct.
- Example: For , arrangements like "A,B,C,D" and "C,D,A,B" are considered different.
- If question has 5 people/object and has circular arrangement the Factorial will be: 5!

2. Circular Arrangement:

- In a circular arrangement, rotating the entire arrangement doesn't create a new arrangement.
 - Example: For "A,B,C,D" and "C,D,A,B is the same.
 - If question has 5 people/object and has circular arrangement the Factorial will be: 4!
-

Probability: memorize these things



Sums on Probability

1. If you throw a coin, find the probability of getting a head?

Ans: Here,

$$S = \{H, T\}; E = \{H\}$$
$$\therefore P(E) = \frac{n(E)}{n(S)}$$
$$= \frac{1}{2}$$

$\therefore$ The probability of getting a head = $\frac{1}{2}$

2. If you throw two coins, find the probability of getting a head?

Ans: Here, $S = \{HH, TT, HT, TH\}$ $E = \{HH, TH, HT\}$

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}$$

$\therefore$ The probability of getting a head is $\frac{3}{4}$

3. Bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is probability that none of the balls drawn is blue?

Ans: Total number of balls:

2 red balls

3 green balls

2 blue balls

$$\therefore n(S) = 7$$

1. Total ways to draw 2 balls

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{7!}{2!(7-2)!} = 21$$

2. Number of ways to draw two ~~blue~~ balls that are not blue:

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{5!}{2!(5-2)!} = 10$$

Let the probability that none of the two balls picked is blue be $P(E)$

$$P(E) = \frac{n(\text{Non-blue ways})}{n(\text{Total ways})} = \frac{10}{21}$$

$\therefore$ The probability that none of the balls drawn is blue is $\frac{10}{21}$

4. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card?

Ans: Total number of cards in the deck is = 52

Total number of face cards $n(F)$ is = 12

Let the probability of drawing a face card be $P(E)$.

$$P(E) = \frac{n(F)}{n(S)} = \frac{12}{52} = \frac{3}{13}$$

$\therefore$ The probability of drawing a face card is $\frac{3}{13}$

5. A card is drawn from a pack of 52 cards. The probability of getting a queen of club or a king of heart is:

Ans: Here, $S = \{52\}$, $E = \{2\}$

$$P(E) = \frac{n(E)}{n(S)} = \frac{2}{52} = \frac{1}{26}$$

$\therefore$ The probability of drawing either the queen of clubs or king of heart is $\frac{1}{26}$

Note:

You may notice that in some probability problems, like "1, 2, 5," I have directly written the sample set and event of interest set as $S\{\}$ and $E\{\}$. In other problems, I have written complete sentences.

Both approaches are correct.

If you have ample time and space left in the exam, I suggest writing out full sentences. However, if you have less time and space, you can use the shortcut method.

Important sums

1. Age & Ratio Problems

1. The ratio of the ages of father and son at present is 6:1. After 5 years the ratio will become 7:2. Find the present age of the son.
 2. Rajeev's age after 15 years will be 5 times his age 5 years back. Find the present age of Rajeev.
 3. Father is twice as old as his daughter. If 20 years ago, the age of the father was 10 times the age of daughter, find the present age of father.
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2. Work & Time Problems

1. P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If both P and Q work together working 8 hours a day, in how many days can they complete the work?
2. A can finish a work in 18 days and B can do the same work in half the time taken by A. Then, working together, what part of the same work they can finish in a day?
3. A does a work in 10 days and B does the same work in 15 days. In how many days will they together do the same work?
4. Two pipes A and B can fill a tank in 36 and 45 hours respectively. If both the pipes are opened simultaneously, how much time will it take to fill the tank?
5. A and B can complete a piece of work in 4 days. If A alone can complete the same work in 12 days, how many days will B require to complete the work alone?
6. Pipe A can fill a tank in 5 hours, pipe B in 10 hours, and pipe C in 30 hours. If all pipes are open, how many hours will it require for the tank to be filled?
7. Two pipes A and B can fill a tank in 20 and 30 minutes respectively. If both pipes are used together, then how long will it take to fill the tank?
8. 39 persons can repair a road in 12 days, working 5 hours a day. In how many days will 30 persons, working 6 hours a day, complete the work?
9. 3 pumps, working 8 hours a day, can empty a tank in 2 days. How many hours a day must 4 pumps work to empty the tank in 1 day?

3. Speed, Distance & Train Problems

1. A train running at a speed of 60 km/hr crosses a pole in 9 seconds. Find the length of the train.
 2. Time is taken by two trains running in opposite directions to cross a man standing on the platform in 28 seconds and 18 seconds respectively. It took 26 seconds for the trains to cross each other. Find the ratio of their speeds.
 3. A person can swim in water with a speed of 13 km/hr in still water. If the stream of the water is 4 km/hr, what will be the time taken by the person to reach 68 km downstream?
 4. In one hour, a boat goes 13 km/hr in the direction of the stream and 7 km/hr against the direction of the stream. What will be the speed of boat in still water?
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4. Probability Problems

1. A bag contains 2 red, 3 green, and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?
 2. If you throw a coin, find the probability of getting a head.
 3. In a simultaneous throw of two coins, find the probability of getting at least one head.
 4. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card?
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5. Profit, Loss & Investment Problems

1. If a bag is purchased for Rs. 500 and sold for Rs. 400, find the loss percentage.
2. Anand and Deepak started a business investing Rs. 22,500 and Rs. 35,000 respectively. Out of a total profit of Rs. 13,800, find Deepak's share.
3. A trader owes a merchant Rs. 10,028 due 1 year hence. The trader wants to settle the account after 3 months. If the rate of interest is 12% per annum, how much cash should he pay?
4. A man buys a watch for Rs. 1950 in cash and sells it for Rs. 2200 at a credit of 1 year. If the rate of interest is 10% per annum, find whether the man gains or loses.

6. Simple & Compound Interest Problems

1. The compound interest on Rs. 30,000 at 7% per annum is Rs. 4,347. Find the period (in years).
 2. Formula to calculate simple interest?
 3. At what rate of compound interest per annum will a sum of Rs. 1,200 become Rs. 1,348.32 in 2 years?
 4. Find the compound interest on Rs. 12,600 for 2 years at 10% per annum, compounded annually.
-

7. Numbers & Divisibility Problems

1. If the number $517*324$ is divisible by 3, find the number in place of *.
 2. If the number $517*324$ is completely divisible by 3, find the smallest whole number in the place of *.
 3. The total population of a city is 6,500. If the number of males and females increases by 5% and 10% respectively, and the total population becomes 7,000, find the number of males in the city.
-

8. Geometry & Measurement Problems

1. A hall is 15 m long and 12 m broad. If the sum of the areas of the floor and the ceiling equals the sum of the areas of the four walls, find the volume of the hall.
 2. In a shower, 5 cm of rain falls. Calculate the volume of water that falls on 1.5 hectares of ground.
 3. Pie chart distribution of land.
 4. The bar graph given below shows the foreign exchange reserves of a country (in million US\$) from 1991-92 to 1998-99. Answer questions based on this.
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9. Miscellaneous / Others

1. A fruit seller had some apples. He sells 40% of the apples and still has 420 remaining. Find how many apples he had originally.
 2. The ratio of 3 numbers is 3:4:5 and the sum of their squares is 1,250. Find the sum of the numbers.
 3. Find H.C.F of 36 and 84.
 4. If 15 toys cost Rs. 234, what is the cost of 35 toys?
 5. Find the permutations of the word "ALLAHABAD".
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How to Write answers in the answersheet

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Q.NO. 4

9) A trader owes a merchant Rs. 10,028 due 1 year hence. The trader wants to settle the amount after 3 months at 12% interest per annum, how much cash to pay?

Ans: Given:

Future Value (FV) = Rs. 10,028
Rate of Interest (i) = 12% per annum
Time (t) = 9 months to 1 year = $\frac{9}{12} = 0.75$ years

Present Value (PV) = $\frac{FV}{1 + (i \times t)}$

= $\frac{10,028}{1 + (0.12 \times 0.75)}$

= $\frac{10,028}{1.09}$

≈ 9200

∴ The trader should pay Rs. 9,200 to settle the amount of account after 3 months.

∴ The correct option is B) 9200

1. You have to solve the entire answer like this and then at the end write what the correct option is like shown in the image above.
2. Even if the question is very simple like i.e "what is the smallest prime number?" then for such small questions you can write the answer in the following way:
The smallest prime number is 2.
Therefore, the correct option is B.
3. Since our gareeb university does not gives us supplements during exams it is wise to use only one page for answer and avoid writing questions.
4. Calculators are allowed in the exam.

